

5. NUTRITION IN PLANTS AND ANIMALS.

Definition.

- Nutrition is the process by which organisms **obtain/acquire and utilize nutrients.**

Importance of nutrition.

- It helps organisms to acquire and utilize nutrients for metabolic activities for respiration, growth and repair of worn out tissues.

Types/modes of nutrition.

1. **Autotrophism-** this is the process through which organisms manufacture/make their own food from simple substances e.g. Carbon (IV) oxide, water in the presence of light/ chemical energy.
 - ✓ The organisms that make their own food are called **autotrophs e.g. green plants, algae and some bacteria.**

2. **Heterotrophism-** this is the mode of nutrition where organisms feed on already manufactured food/ complex food materials e.g. carbohydrates, proteins and lipids (fats and oils).
- ✓ Organisms that feed on already manufactured food are called heterotrophs e.g. animals, protozoa (e.g. amoeba) and some bacteria.

NUTRITION IN PLANTS

- ✓ Nutrition in plants is called autotrophism.

Types/ modes of autotrophism

1. **Photosynthesis-** this is the process by which green plants manufacture their own food using light energy.
- ✓ During photosynthesis light energy is converted into chemical energy and stored in food.
2. **Chemosynthesis-** this is the process through which organisms (e.g. bacteria) manufacture their own using chemical energy obtained from chemical reactions.

External parts of a dicot leaf.

1. **Leaf lamina/blade-** it is flat and broad to increase surface area for trapping light energy for photosynthesis. The lamina is green in color and contains the photosynthetic tissue.
2. **Mid rib and veins-** the mid rib is thick and runs in the middle from the petiole to the apex. The veins run into the lamina forming an extensive network of **veins**. The veins have:
 - i) **Xylem-** which transports water and mineral salts to the photosynthetic cells.
 - ii) **Phloem-** which transports manufactured food from photosynthetic cells.
3. **Petiole-** Attaches the leaf to the branch or stem.
4. **Margin-** it is either smooth or serrated.
5. **Apex-** It is located at the tip of the blade.

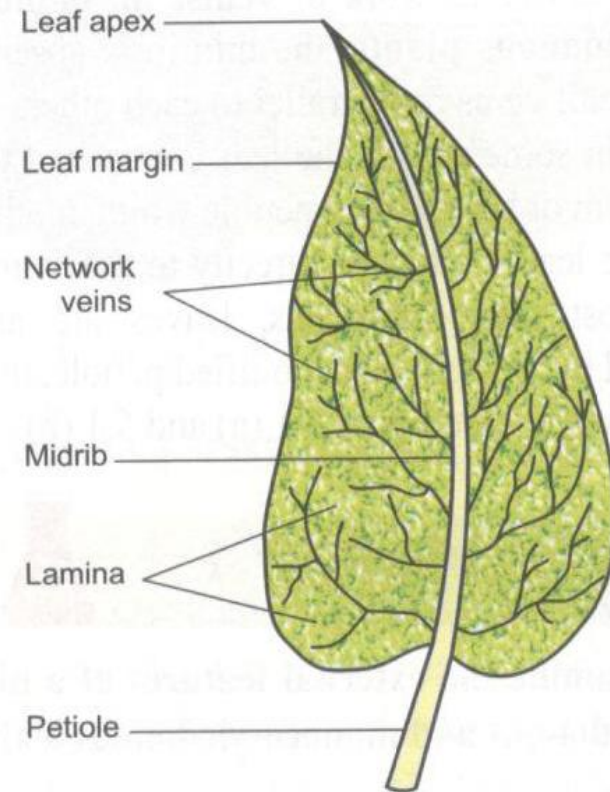
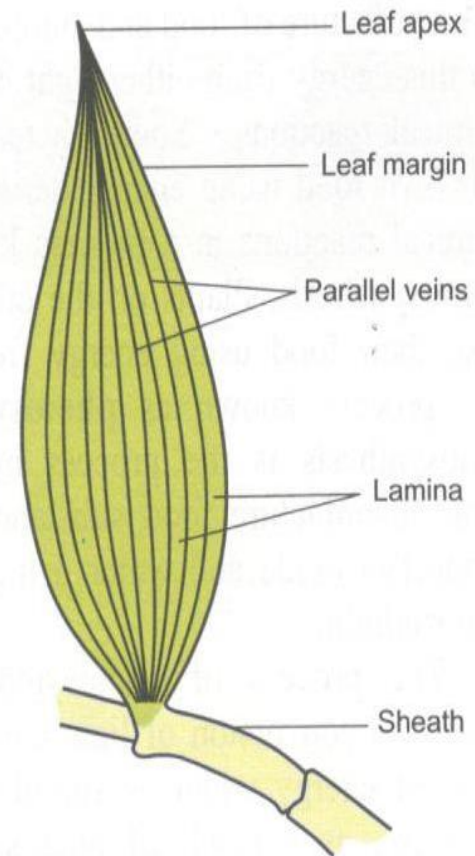


Fig 5.1 (a): External features of a dicotyledonous leaf, e.g., bean.

EXTERNAL PARTS OF A MONOCOT LEAF.

1. **Leaf lamina/blade-** it is flat and narrow to trap light energy for photosynthesis. The lamina is green in color and contains the photosynthetic tissue.
2. **Parallel veins-** they run parallel from sheath to apex. The veins have:
 - i) **Xylem-** which transports water and mineral salts to the photosynthetic cells.
 - ii) **Phloem-** which transports manufactured food from photosynthetic cells.
3. **Margin-** it is either smooth or serrated.
4. **Apex-** It is located at the tip of the blade.
5. **Leaf sheath-** it attaches the leaf to the stem/ branch.



(b): External features of monocotyledonous leaves, e.g. Zebrina.

Differences between dicot and monocot leaf

Dicot leaf

1. Has broad lamina/ blade.
2. Has leaf petiole.
3. Has midrib.
4. Has network veins.

Monocot leaf

1. Has narrow lamina/ blade.
2. Has leaf sheath.
3. Lacks midrib.
4. Has parallel veins.

INTERNAL STRUCTURE OF A LEAF.

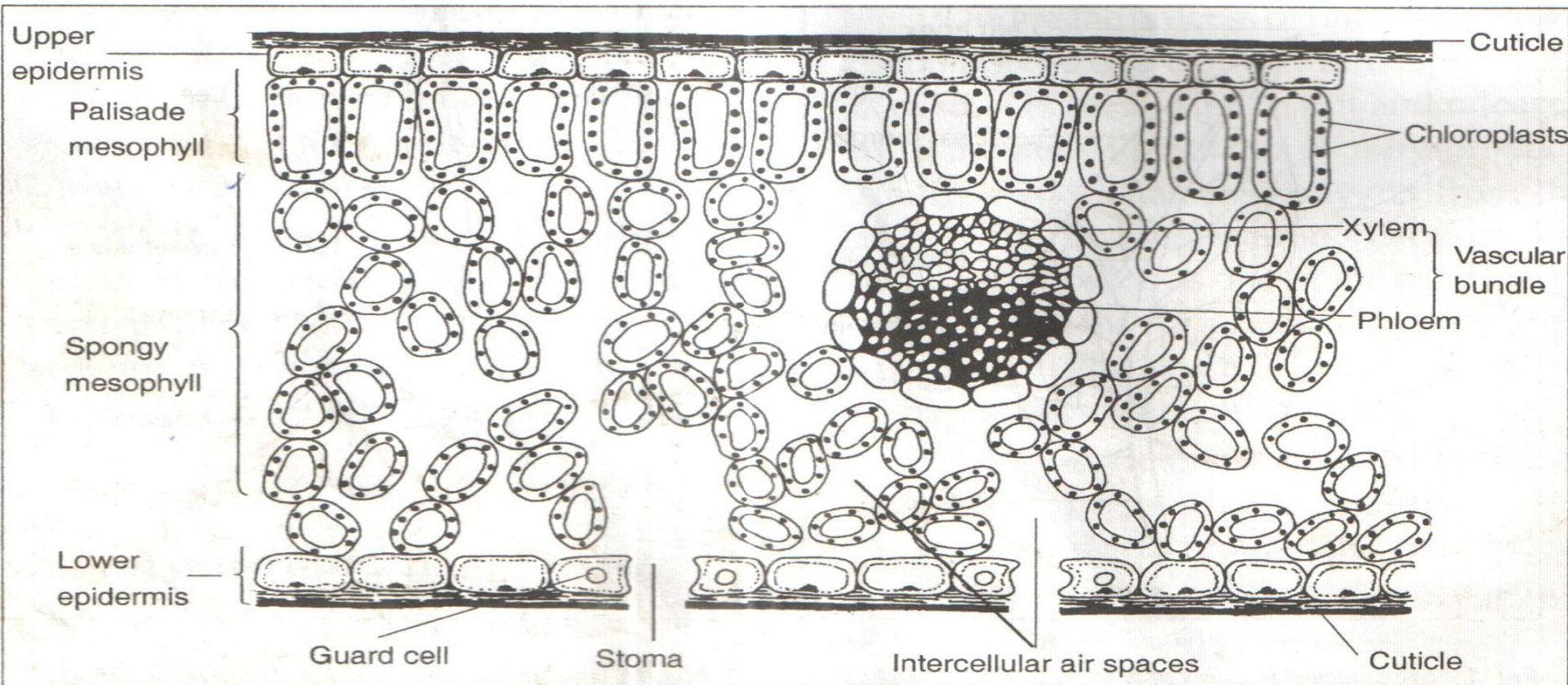


Fig. 5.2: Transverse section of a leaf

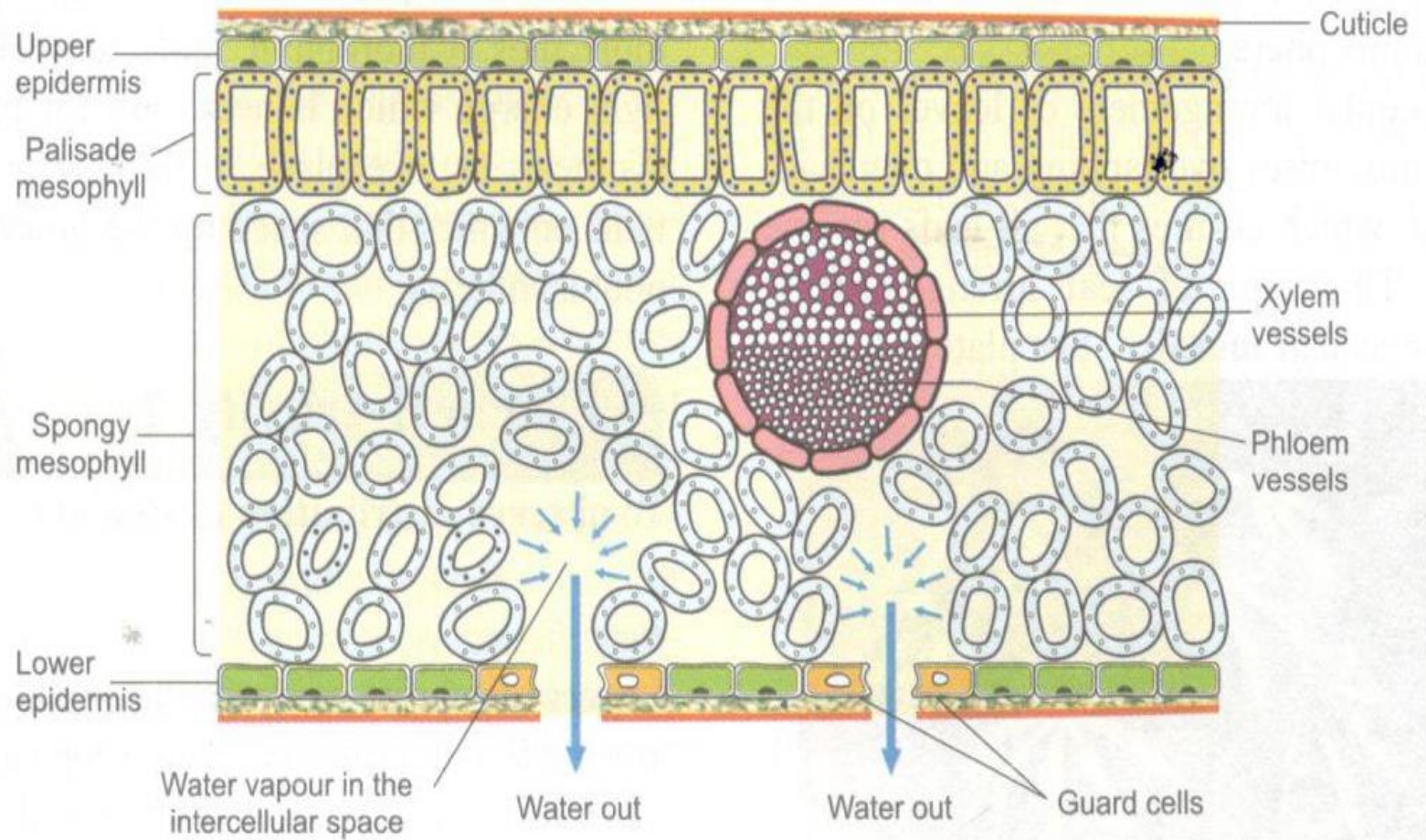


Fig 5.2: Transverse section of a leaf.

Internal parts of a leaf

1. **Cuticle**- This is a thin non-cellular, waxy, waterproof and transparent layer that covers the upper and lower surfaces of the leaf. The cuticle is transparent to allow penetration of light for photosynthesis.

Functions of cuticle

- i. It reduces excessive loss of water.
- ii. Protects the inner tissues of the leaf from mechanical damage.
- iii. Prevents entry of disease-causing microorganisms.

2. **Epidermis**- This is a thin tissue, usually one cell thick, on the upper and lower surfaces of the leaf. The epidermal cells have no chloroplasts *except the guard cells*. Chloroplasts are the organelles that contain the photosynthetic material known as **chlorophyll**.

Functions/ adaptations of epidermis.

- a) It secretes the cuticle.
- b) It is transparent to allow light penetrate to the photosynthetic tissue.
- c) It has a single layer of cell (i.e. thin) to reduce the distance over which light penetrates to photosynthetic tissue and carbon (IV) oxide diffuses to photosynthetic cells.
- d) It has stomata for gaseous exchange.
- e) There are guard cells to control opening and closing of stomata.
- f) It is covered with thick, waxy and has waterproof cuticle to prevent excessive water loss and protect the inner parts of the leaf.

3. **Guard cells**– they are specialized epidermal cells used to control opening and closing of stomata.

- Unlike the epidermal cells, the guard cells are bean-shaped while other epidermal cells are blocky shaped. They also contain chloroplasts/ are able to carry out photosynthesis

Structural adaptations of guard cells.

- a) They have thicker inner less elastic walls which curve to open the stomata and straighten to close the stoma.
- b) They have outer thinner and less elastic walls which bulge outwards.
- c) They contain chloroplasts for photosynthesis/ manufacture glucose which is osmotically active.

4. **Palisade layer/ cells-** This is a layer of cells located beneath/below the upper epidermis. It consists of cylindrical shaped cells closely packed together and with the long axis perpendicular to the surface.

Adaptation to function

- a) They have numerous chloroplasts containing chlorophyll which is necessary for photosynthesis. Their position and arrangement enables them to receive maximum sunlight.

5. **Spongy mesophyll layer-** this is a layer of cells between the palisade and the lower epidermis. The cells are irregularly shaped and loosely arranged creating large air spaces in between them.

- ✓ The air spaces provide communication pathways through which gases diffuse in between the cells.
 - ✓ Unlike the palisade cells, spongy mesophyll cells contain fewer chloroplasts.
 - ✓ This explains why the lower surface of the leaf is lighter in colour than the upper surface.
6. **Vascular bundle-** consists of xylem and phloem tissues.
- ✓ The xylem transport water and mineral salts from the roots to the leaf cells.
 - ✓ The phloem transports/translocates manufactured food from the leaf cells to the rest of the plant.

Study question

Name three cells in a leaf which contain chloroplasts.

- i. **Palisade.**
- ii. **Spongy mesophyll.**
- iii. **Guard cells.**

ADAPTATIONS OF THE LEAF TO PHOTOSYNTHESIS.

1. It is green in colour/ contain chlorophyll which traps sunlight energy needed for photosynthesis.
2. Has broad and flat lamina which provides a large surface area for the absorption of carbon (IV) oxide trapping sunlight.
3. It has thin lamina to allow light and carbon (IV) oxide to pass through a short distance to reach the photosynthetic cells.
4. It has stomata to ensure efficient diffusion of respiratory gases in and out of the leaf.
5. It contains guard cells which control opening and closing of stomata; and contain chloroplasts/ chlorophyll to trap light energy/ carry out photosynthesis.
6. It has transparent cuticle and epidermis to allow penetration of light to the palisade cells.
7. The palisade cells and mesophyll cells contain large numbers of chloroplasts located next to the upper epidermis enables them to receive maximum sunlight.
8. The mid-rib and veins contain xylem which transports water and mineral salts to photosynthetic cells and phloem; which transports manufactured/ photosynthetic materials from photosynthetic cells
9. Spongy mesophyll cells have large air spaces which allow circulation of air to facilitate gaseous exchange.
10. The regular /mosaic arrangement of leaves on the stem minimizes overlapping and overshadowing so that each leaf receives adequate light
11. Has xylem, parenchyma and sclerenchyma tissues to support the leaf exposing it to sunlight.

Study questions

1. List three adaptations of leaves that maximize efficiency in trapping sunlight for photosynthesis.
 - i. **Flat broad lamina.**
 - ii. **Transparent cuticle.**
 - iii. **Thinness of the leaf.**
2. Explain how aquatic plants are adapted to photosynthesis.
 - *Emergent and floating hydrophytes have broad leaves with numerous stomata* on the upper surface to increase the surface area for transpiration and for efficient gaseous exchange.
 - *Submerged hydrophytes have highly dissected leaves into thread-like straws* to increase surface area for absorption of maximum light and CO_2 for photosynthesis and gaseous exchange.
 - The leaves of *Submerged hydrophytes have numerous and sensitive chloroplasts* that photosynthesize under low light intensities.



Plate 5.1: Leaf mosaic in cassava

THE CHLOROPLAST.

Parts of chloroplast

1. The outer and inner membranes.
2. Inner membrane is folded to form **lamellae** suspended in an aqueous matrix called **stroma**.
3. The lamellae may at certain intervals form several layers of membranes grouped together to make a **granum** (plural – grana).
4. **Granum** contains **chlorophyll**- which absorb light energy which is necessary for photosynthesis.
5. **Stroma** contains enzymes that speeds up the process of photosynthesis.

Adaptations of chloroplast to its function (photosynthesis).

1. It has lamellae/grana that contains chlorophyll that traps light energy.
2. The grana has a large surface area for accommodation or packing of the chlorophyll.
3. The stroma contains the enzymes that speed up/catalyze the process of photosynthesis.

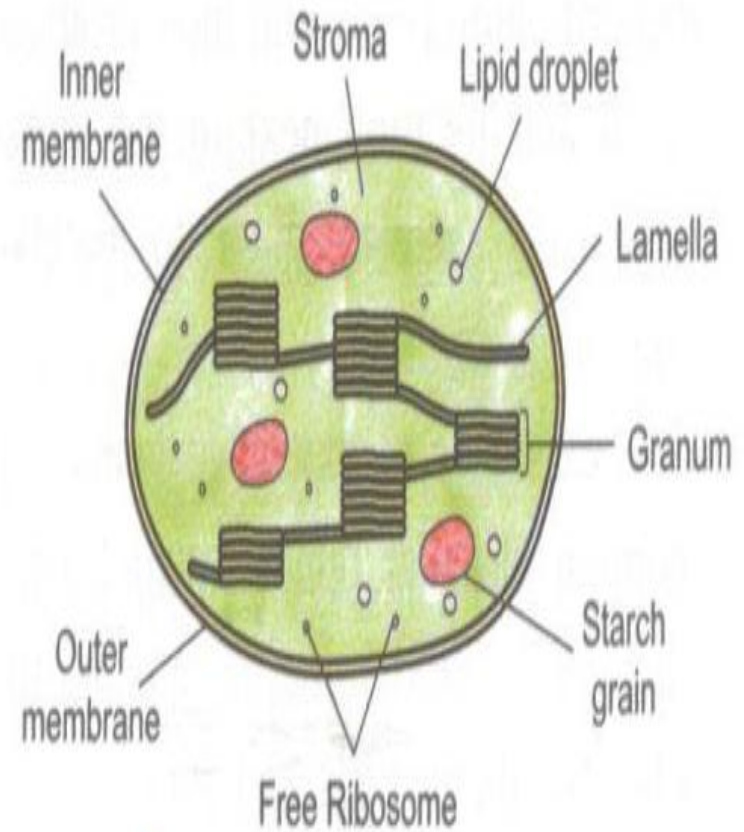
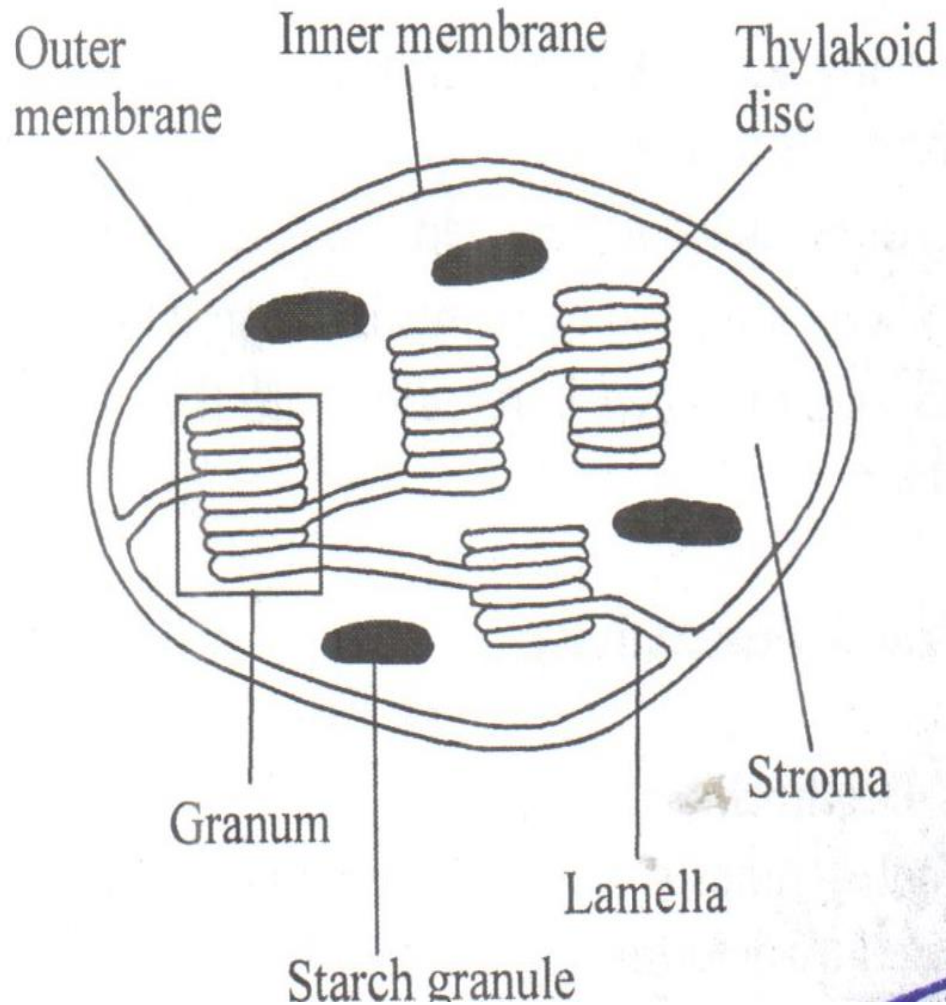


Fig 5.3: Section through a chloroplast

The process of Photosynthesis.

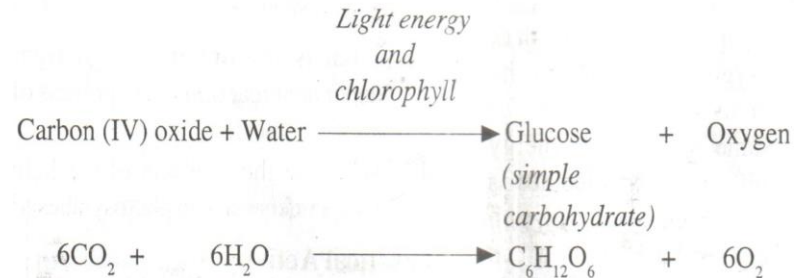
- The raw materials for photosynthesis are:

- i. Water
- ii. Carbon (IV) oxide gas.

Conditions required for photosynthesis to take place

- i. Light and
- ii. Chlorophyll

- *Water and carbon (IV) oxide* molecules undergo several chemical processes in the presence of sunlight to form carbohydrates / glucose, oxygen and energy (ATP) are given out as a by-product.



Importance of photosynthesis.

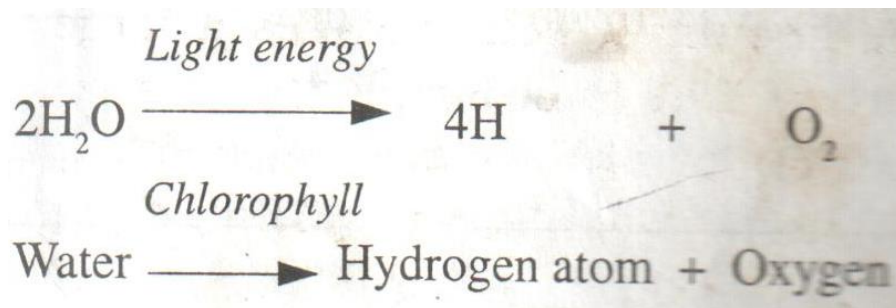
1. Acts as a source of energy.
2. It provides oxygen in the air.
3. It prevents the accumulation of carbon (IV) oxide in the atmosphere.

Stages of photosynthesis.

- A. Light stage/ light dependent stage.
- B. Dark stage/ light independent stage/ Carbon (IV) oxide fixation.

A. Light stage (light dependent stage).

- This is the initial stage in the process of photosynthesis.
- It occurs in the **grana** of the chloroplasts.
- The chlorophyll molecule absorbs/traps light energy which **is used to split water molecules into oxygen and hydrogen atoms.**
- This process is called **photolysis** of water.
- The **hydrogen** atoms that are produced enter the dark stage.



- **Oxygen** is released to the atmosphere or used by the plant for respiration.
- **Adenosine Triphosphate (ATP)** is formed by energy light which is later used in the dark stage.
- This reaction involves conversion of light energy to chemical energy.

Importance of light stage.

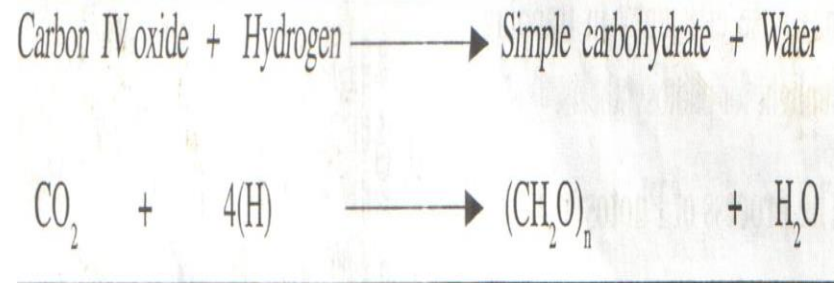
- It provides hydrogen atoms and ATP molecules useful during carbon (IV) oxide fixation.

Fate of end products of light stage. (what happens to the end products of light stage?)

- i) Hydrogen atoms enter the dark stage.
- ii) Oxygen atoms are released to the atmosphere as a gas or used for respiration.

B. The Dark Stage/carbon (IV) oxide fixation/ (Light Independent Stage).

- It occurs in the **stroma** in the presence or absence of light.
- It involves the combination of **carbon (IV) oxide** with **hydrogen atoms** to form simple sugar /*glucose, fatty acids and amino acids*.
- The energy required for this reaction is provided by ATP from light stage reaction.
- Glucose formed is converted into starch for storage. *This is important because starch is osmotically inactive (compared to glucose) hence the plant cells do not lose water through osmosis.*
- If the leaf is kept in darkness for 48 hours, starch is converted into glucose. This is called *destarching*.



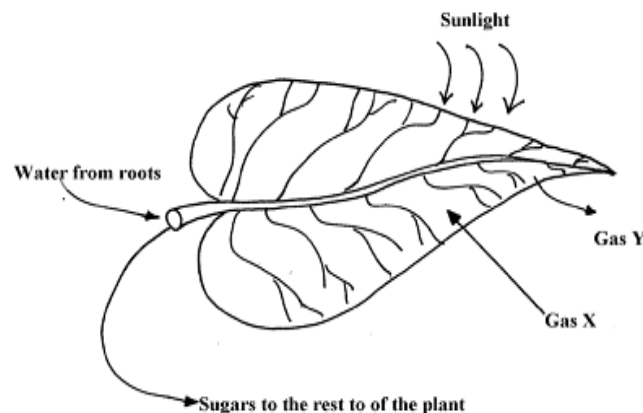
What happens to end products of photosynthesis?/ the Fate of end products of photosynthesis.

- ✓ Some glucose is used in respiration.
- ✓ Some glucose is converted into starch for storage.
- ✓ Some glucose is converted into sucrose which is translocated to other parts of the plant.
- ✓ Some glucose is used in making cellulose for the cell wall.
- ✓ Fatty acids and glycerol are combined to form oils and fats (lipids).
- ✓ Amino acids are converted to proteins.
- ✓ Oxygen is used by plants in respiration.
- ✓ Excess oxygen is released into the atmosphere.

Study questions.

1. State four requirements and sources for the process of photosynthesis to occur.
 - ✓ *Water- soil.*
 - ✓ *Carbon (IV) oxide- atmosphere.*
 - ✓ *Chlorophyll- available on the chloroplasts.*
 - ✓ *Light- sunlight*
2. Give the role of each of the following structures in the leaf during photosynthesis.
 - a) Xylem vessels- *transports water from the soil to the leaves where photosynthesis takes place.*
 - b) Chlorophyll- *traps light energy (used for photolysis).*
 - c) Guard cells- *control opening and closing of stomata which allow in carbon (IV) oxide for use during photosynthesis.*

3. Explain why plants will not photosynthesise in the dark.
- ✓ ***Photosynthesis is the process by which green plants manufacture food in the presence of light hence it does not take place in the absence of light.***
4. Explain why most leaves are thin with broad surface.
- ✓ ***They are thin and broad to provide a large surface area for absorption of light and carbon (IV) oxide for maximum photosynthesis.***
5. The following diagram of a leaf shows what happens in a plant leaf during photosynthesis.



- a) Name the gases labelled **X** and **Y**.
 - X – Carbon (IV) Oxide.
 - Y – Oxygen.
- b) Give **two** ways in which leaves are adapted to absorb light.
 - ***Broad and flat to absorb maximum light.***
 - ***Have chloroplast with chlorophyll to trap light.***
 - ***Transparent cuticle to allow light to pass through.***
- c) Name the tissue that transports water into the leaf and sugars out of the leaf.
 - ***Xylem – Transports water.***
 - ***Phloem – Sugars out of the leaf.***
- d) Explain why it's an advantage for the plant to store carbohydrates as starch rather than as sugars.
 - ***Starch is insoluble in water, hence osmotically inactive. This reduces effect on absorption of water by cells.***

6. A group of students placed a fresh leaf in warm water. They observed that air bubbles formed on the surface of the leaf.

a) What biological process were they investigating?

Photosynthesis.

b) Name the structures from which the air bubbles were coming from.

Stomata.

c) Explain the distribution of the structures named in (b) above on the leaf surfaces.

They are more on the lower surface and fewer on the upper surface of terrestrial plants to reduce the rate of transpiration.

7. **Explain the formation of starch in green plants.**

- *Green plants manufacture food through the process of photosynthesis; the leaves contain chloroplasts which contain chlorophyll where photosynthesis take place;*
- *Light stage occurs in the granum; where chlorophyll traps light energy; which splits water into hydrogen ions and oxygen atoms/ photolysis; and ATP/ energy used in dark stage;*
- *Dark stage occurs in the stroma; where carbon (IV) oxide from the atmosphere; combines with hydrogen atoms to form simple sugars/ glucose molecules; which are converted into starch for storage;*

8. Other than photosynthesis, explain how carnivorous/ insectivorous plants obtain nutrients.

- ✓ *They grow in nitrogen deficient soil and obtain nitrogen from insects.*
- ✓ *Insects are attracted by colour/ scent/ sugary baits and are trapped by plant (nastic responses) and digested by proteases secreted by insects.*

FACTORS AFFECTING THE RATE OF PHOTOSYNTHESIS.

a) Environmental/ external factors

1. Light intensity and quality.
2. Carbon (IV) oxide concentration.
3. Temperature.
4. Water.

b) Internal factors

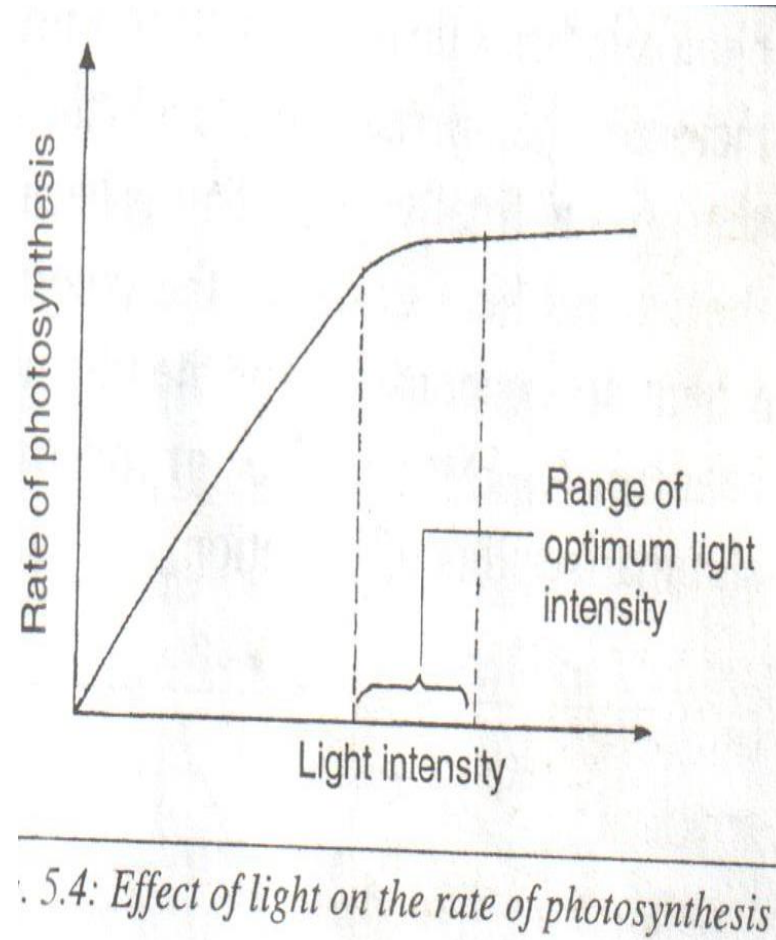
1. Chlorophyll concentration.
2. Enzyme concentration.

1. LIGHT INTENSITY / QUALITY OF LIGHT

- ✓ Light provides the energy required for the process of photosynthesis.
- ✓ Decrease in light intensity decreases the rate of photosynthesis.
- ✓ The rate of photosynthesis increases as light intensity increases up to an optimum level.
- ✓ Beyond the optimum level the rate of photosynthesis remains constant due to other limiting factors.
- ✓ At very high light intensity, chlorophyll is damaged/ bleached and the rate of photosynthesis falls.

Note.

- There is no grass/ vegetation in dense forests because forests form canopies/ shadow which prevents light from reaching grass/ vegetation thus grass/ vegetation die/ fail to flourish due to inactivity to photosynthesize.



2. CARBON (IV) OXIDE CONCENTRATION.

- ✓ Carbon (IV) oxide is a raw material for photosynthesis.
- ✓ Decrease in carbon (IV) oxide decreases the rate of photosynthesis.
- ✓ An increase in carbon (IV) oxide (beyond 0.04%) leads to an increase in the rate of photosynthesis up to a given optimum.
- ✓ Beyond optimum, the rate of photosynthesis remains constant due to other limiting factors.

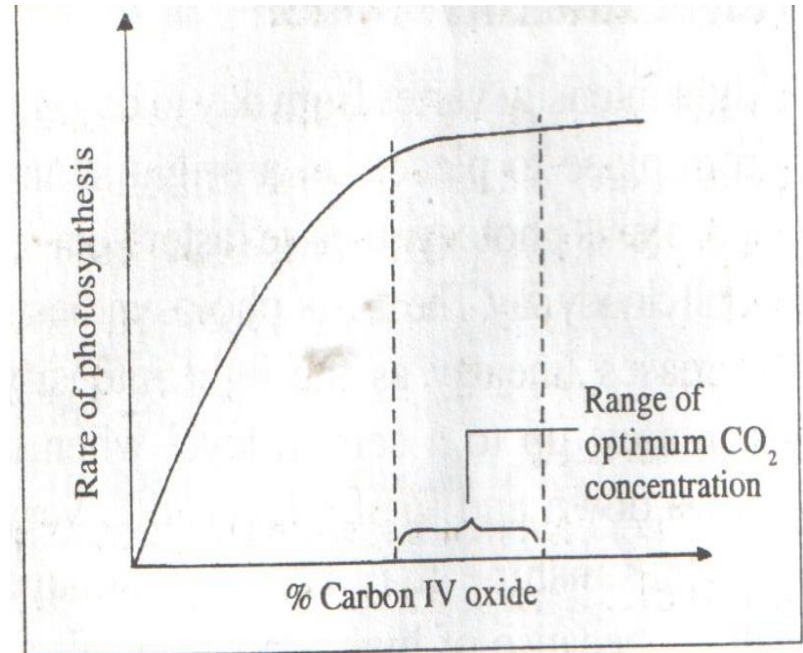
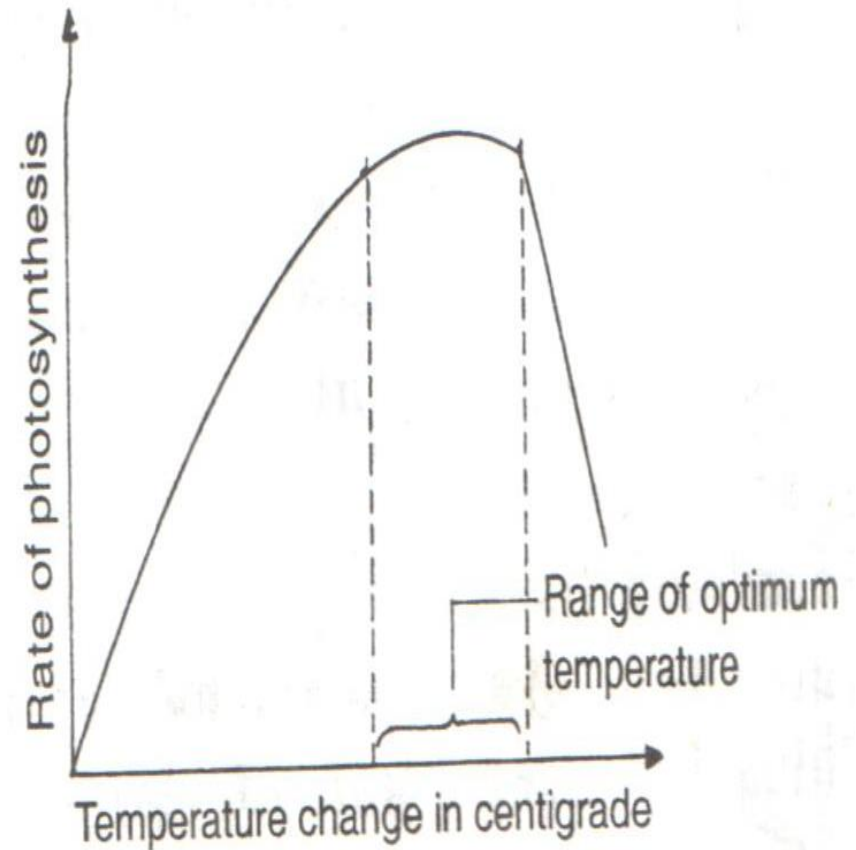


Fig. 5.5: Effect of carbon (IV) oxide on the rate of photosynthesis

3. TEMPERATURE.

- ✓ Photosynthesis reactions are catalyzed/controlled by enzymes.
- ✓ Decrease in temperature decreases the rate of photosynthesis because enzymes are inactivated.
- ✓ Increase in temperature activates enzymes thus increasing the rate of photosynthesis up to optimum.
- ✓ High temperatures above optimum denatures enzymes, thus reducing the rate of photosynthesis.



4. WATER.

- ✓ Water is a raw material for photosynthesis.
- ✓ It influences opening and closure of stomata which affects the diffusion of carbon (IV) oxide into the leaf thus further affecting the rate of photosynthesis.

5. CHLOROPHYLL CONCENTRATION.

- ✓ Chlorophyll is the pigment that traps light energy during photosynthesis.
- ✓ The higher the chlorophyll concentration the higher the rate of photosynthesis and vice versa.
- ✓ A **variegated leaf** is one that has some patches that lack chlorophyll. These patches have other colours e.g. yellow.

- ✓ These parts lack chlorophyll hence do not photosynthesize and therefore gives negative results with starch test.
- ✓ The variegated leaf has less starch than a normal leaf **because it has less chlorophyll hence manufactures less food.**

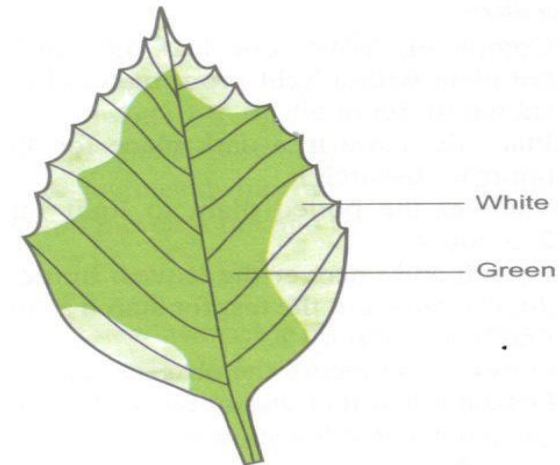


Fig 5.10: A variegated leaf.

6. ENZYME CONCENTRATION

- ✓ Enzymes catalyze the photosynthetic reactions.
- ✓ Increase in enzyme concentration increases the rate of photosynthesis.
- ✓ A decrease in enzyme concentration decreases the rate of photosynthesis.

Study question

1. Variegated plants accumulate less food than non-variegated plants under similar conditions.
explain
 - ✓ *Variegated leaves have less chlorophyll compared to non-variegated leaves. They absorb less light hence facilitate less photosynthesis.*

PRACTICAL ACTIVITY 1.

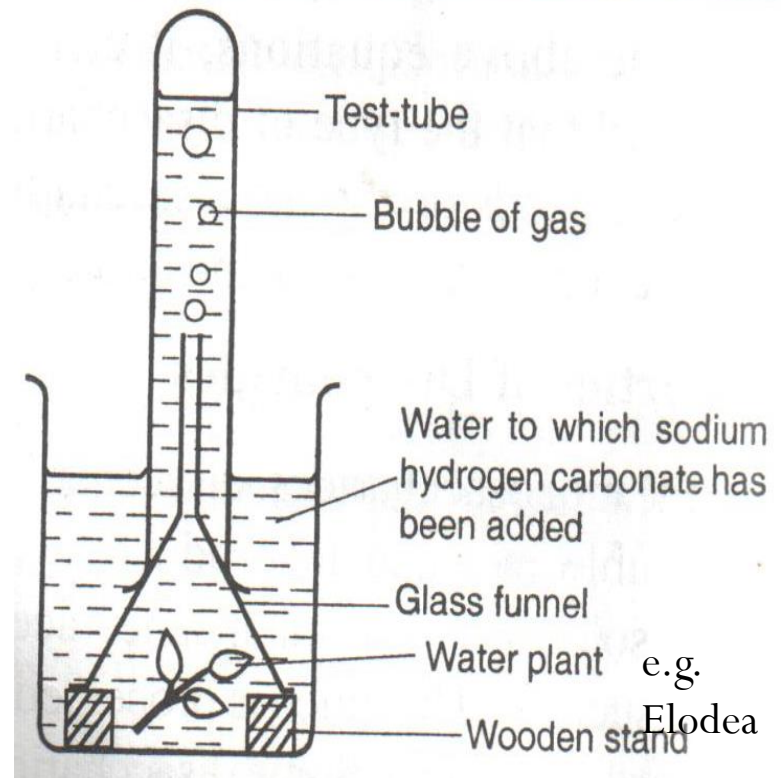
Aim: To investigate the gas produced during photosynthesis.

Requirements:

1. Water plant, e.g. *Elodea* spp., Spirogyra or Nymphaea (water lily)
2. Glass funnel.
3. Beaker.
4. Small wooden blocks.
5. Test tube.
6. Wooden splint.
7. Sodium hydrogen carbonate.

Procedure

1. Set-up the experiment as shown below.
2. Place the set-up in the sunlight to allow photosynthesis to take place.
3. Leave the set-up in the sun until sufficient gas has collected in the test-tube.
4. Test the gas collected with a glowing splint.



Questions.

1. What gas is produced during photosynthesis?
 - ***Oxygen.***
2. Why was sodium hydrogen carbonate used during the experiment?
 - ***To increase the amount of carbon (IV) oxide in water and accelerates the rate of photosynthesis***
3. Explain why only submerged water plants are used instead of terrestrial plants in the experiment.
 - ***Submerged plants are adapted to aquatic conditions hence can carry out photosynthesis in water.***
4. Name other factors that can be tested using the set up above.
 - ✓ Temperature.
 - ✓ Light intensity.
 - ✓ Carbon (IV) oxide concentration.

PRACTICAL ACTIVITY 2.

Aim: To test / investigate the presence of starch in a leaf.

Requirements:

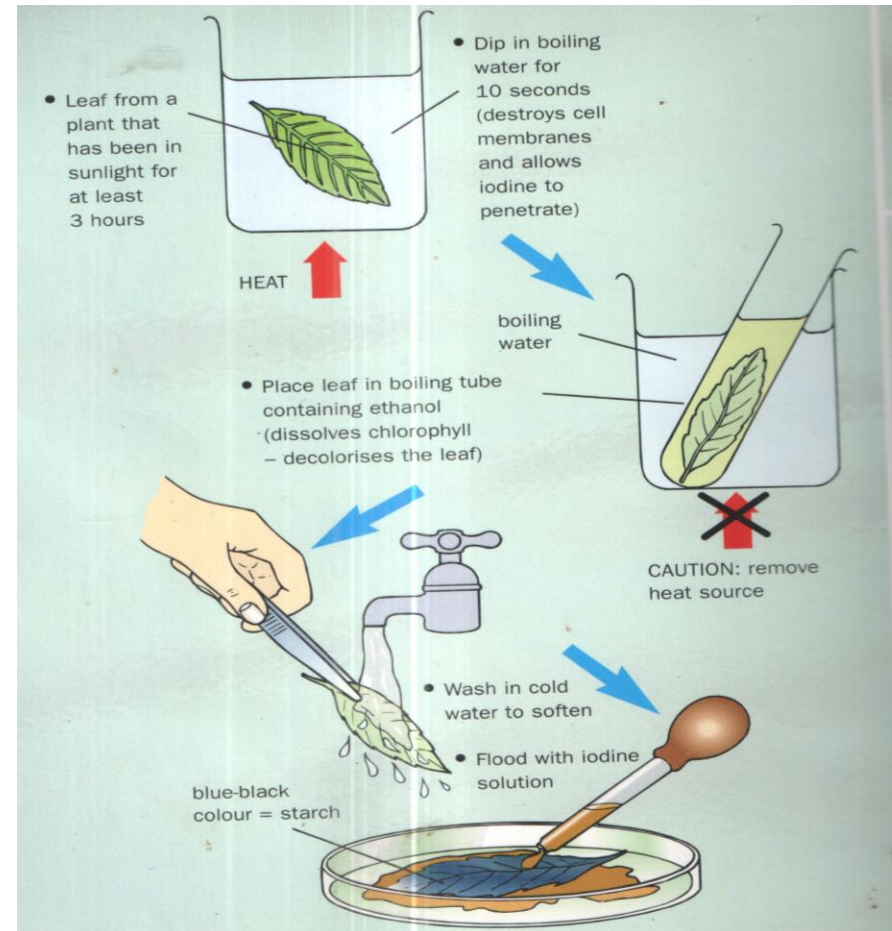
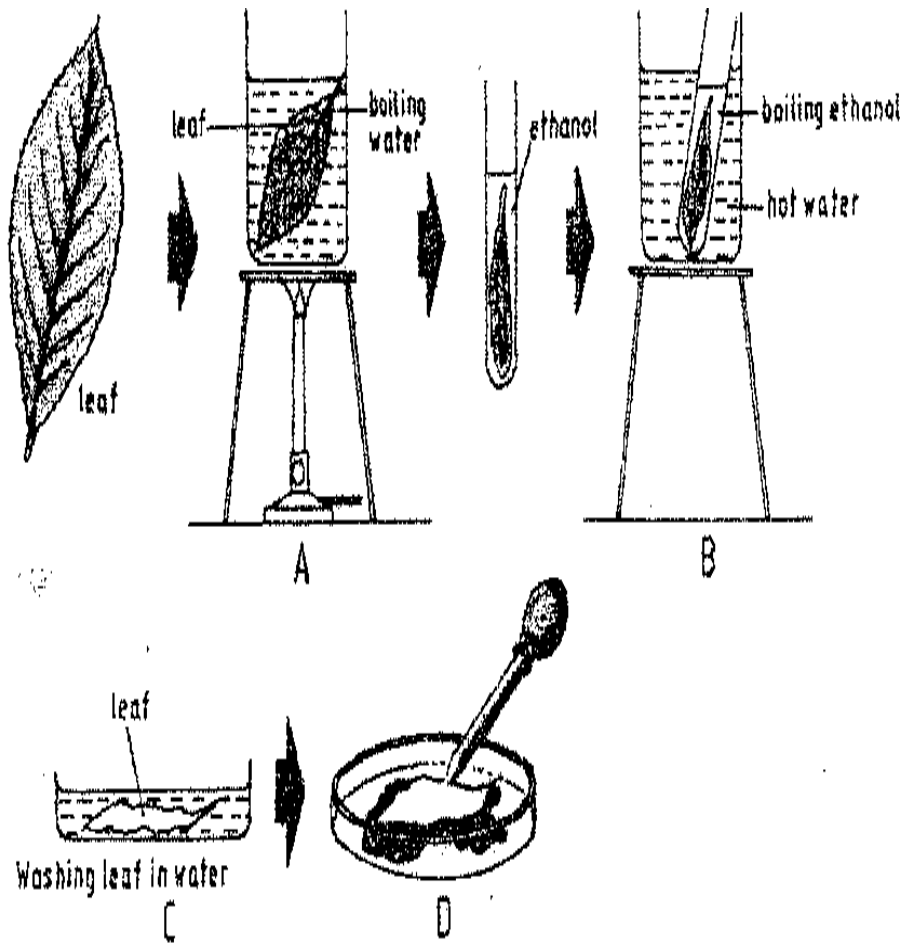
1. Water.
2. Dropper.
3. Beaker.
4. Source of heat.
5. Boiling tube.
6. A normal and variegated leaf.
7. Petri dish.
8. White tile.
9. Iodine solution.
10. Methylated spirit.

Procedure:

1. Obtain a leaf that has been exposed to light for at least 5 hours.
2. Boil water in a beaker.
3. Dip the leaf in the boiling water for 3 to 4 minutes.
4. Put the leaf in a boiling tube containing methylated spirit and stand the tube in the beaker containing boiling water (water bath) for about 10 minutes **to decolorize the leaf**. Avoid direct heating because spirit is highly flammable.
5. Remove the leaf from the test-tube and wash it in warm water in the beaker **to soften it**.
6. Spread the leaf in a Petri-dish and add drops of dilute iodine solution.
7. Observe and record the colour changes.

Observation.

- The colour changes to blue-black indicating the presence of starch.



Questions.

1. Why is it important to use the leaf that has been exposed to light for a few hours?
 - ✓ *To ensure that photosynthesis occurs and starch is formed.*
2. Why is a fresh leaf dipped in boiling water?
 - ✓ *To kill the protoplasm of the cell/kill the leaf cells/breakdown starch granules/ stop enzymatic activity.*
3. Give a reason why the leaf was dipped in ethanol/ methylated spirit.
 - ✓ *To remove chlorophyll/ dissolve chlorophyll /decolorize the leaf.*
4. Why was the leaf decolourised?
 - ✓ *To make the color change in iodine to be seen clearly.*
5. Why is methylated spirit boiled indirectly?
 - ✓ *It is highly flammable.*
6. Why was the leaf dipped in water?
 - ✓ *To soften it.*
7. Explain why a leaf cannot be tested for starch by adding iodine solution directly.
 - ✓ *The leaf must be dipped in water to kill it then boiled in methylated spirit to decolorise it to make color change in iodine to be seen clearly.*

PRACTICAL ACTIVITY 3.

Aim : To investigate whether light is necessary for photosynthesis.

Requirements:

1. Methylated spirit.
2. Iodine solution.
3. Water.
4. White tile.
5. Droppers.
6. Beaker.
7. Source of heat.
8. Boiling tube.
9. Light proof material (e.g. Aluminium foil).
10. Potted plant.
11. Clips.

Procedure.

1. Cover one leaf of a potted plant with a light-proof material as shown below.
2. Place the plant in a dark place for 48 hours **to destarch it/ ensure that all starch has been used up.**
3. Transfer the potted plant to light for 2-3 hours.
4. Detach and uncover the leaves and immediately carry out the test for starch.

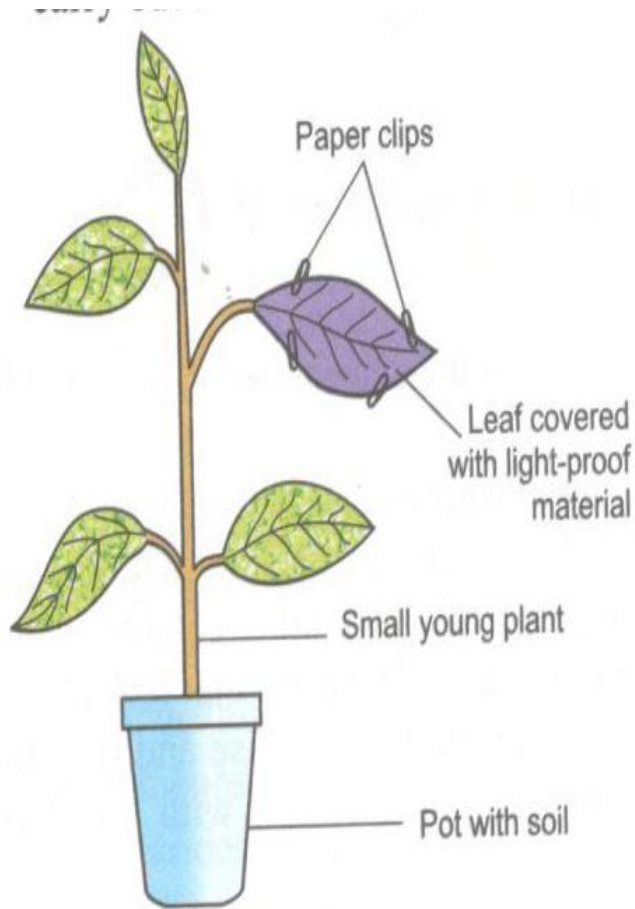


Fig 5.8: Necessity of light in photosynthesis.

Questions

1. Why was the plant kept in the dark for 48 hours?
 - *To destarch it / to ensure that all starch in it is used up.*
2. Why was it necessary to transfer the plant to light?
 - *To allow the plant to photosynthesize and hence manufacture starch.*
3. What was the role of the lightproof paper?
 - *To reflect light so that none is absorbed by the leaves.*

PRACTICAL ACTIVITY 5.

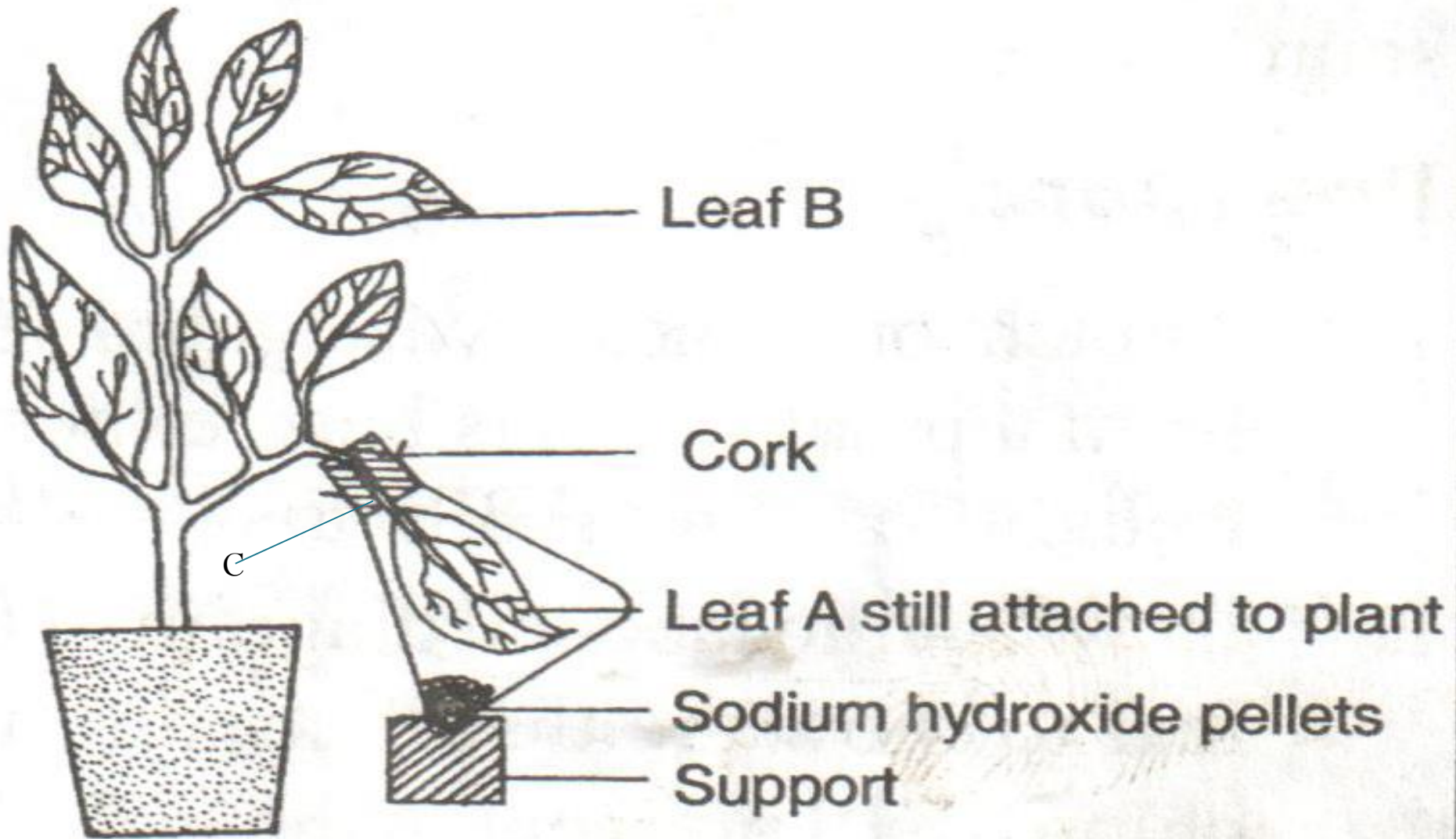
Aim: To investigate whether carbon (IV) oxide is necessary for photosynthesis.

Requirements:

1. Conical flasks/polythene bags.
2. Potted plant.
3. Sodium hydroxide pellets.
4. Cork or plasticine or clay.
5. Cork borers.
6. Scalpel.
7. Petroleum jelly.
8. Iodine solution.
9. Methylated spirit.
10. Water.
11. Beakers.
12. Droppers.
13. White tiles.
14. Boiling tubes.
15. Source of heat.
16. Wooden support.

Procedure.

1. Keep the potted plant in a dark place for 48 hours.
2. Place a few pellets of sodium hydroxide in the flask.
3. Bore a hole in the cork of the same size as the petiole of the leaf being used.
4. Using the scalpel, cut the cork lengthwise.
5. Remove the plant from the dark and immediately fit the petiole of a leaf A in the groove and cork the flask as shown below.
6. Seal the mouth of the conical flask with petroleum jelly to make it airtight.
7. Keep the set up in the light for two to three hours.
8. Detach and test for the presence of starch in both leaves A and B.



Questions

1. What is the function of the **sodium hydroxide** pellets?

To absorb carbon (IV) oxide.

2. Why was the leaf outside the flask also tested for starch?
 - *It was the control experiment.*
3. Giving a reason explain the expected result after testing for starch on the part labeled C.
 - *Starch absent. This is because of the absence of light and carbon (IV) oxide.*

4. Explain the result obtained after testing for the presence of starch in leaves A and B.

- *Starch was present in leaf B. This is because it was exposed to sunlight hence carried out photosynthesis forming starch.*
 - *Starch was absent in leaf A. This is because sodium hydroxide pellets absorbed carbon (IV) oxide which is a raw material for photosynthesis.*
5. What is the expected result for starch in leaf A if **sodium hydrogen carbonate** is used instead of **sodium hydroxide**?
 - *Starch was present in leaf A. This is because sodium hydrogen carbonate breaks down to give carbon (IV) oxide which is a raw material for photosynthesis.*

PRACTICAL ACTIVITY 6.

Aim: To investigate whether chlorophyll is necessary for photosynthesis.

Requirements.

1. Variegated leaves.
2. Iodine solution.
3. Methylated spirit.
4. White tile.
5. Water.
6. Boiling tubes.
7. Beaker.
8. Dropper.
9. Source of heat

Procedure

1. Detach a variegated leaf from the plant that has been exposed to light for at least 3 hours.
2. Draw a large labeled diagram of the leaf to show the distribution of the chlorophyll pigment in the leaf as shown.
3. Test for the presence of starch.

Observation and explanation.

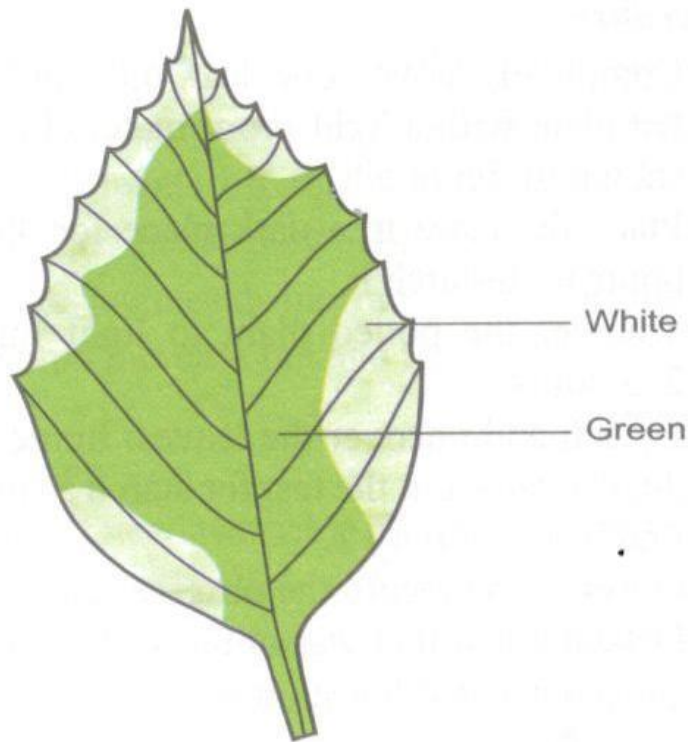


Fig 5.10: A variegated leaf.

- ✓ The green parts/ patches give blue-black color with iodine. This is because they contain chlorophyll hence carry out photosynthesis forming starch.
- ✓ The white parts/ patches retained brown color with iodine. This is because they lack chlorophyll hence did not carry out photosynthesis forming starch.
- ✓ Starch was found on green patches but not on white patches.

CHEMICAL COMPOUNDS WHICH CONSTITUTE LIVING ORGANISMS/ CHEMICALS OF LIFE

- ✓ These are compounds found in cells, tissues and organs.
- ✓ The study chemical compounds found in living organisms and reactions in which they take place is called **biochemistry**.
- ✓ Some of the chemical compounds are organic e.g. carbohydrates, proteins, lipids, nucleic acids and vitamins.
- ✓ Other chemical compounds are inorganic compounds e.g. mineral salts, water, acids and bases.

1. CARBOHYDRATES.

- ✓ They are compounds that contain carbon, hydrogen and oxygen in the ratio of 1 carbon : 2 hydrogen : 1 oxygen.
- ✓ The basic formula is $(\text{CH}_2\text{O})_n$.
- ✓ The constituents of carbohydrates are joined by a bond called **glycosidic bond**.

Uses/ functions of carbohydrates.

1. They are sources of energy/ are broken down to give energy. Glucose is the main source of energy.
2. They are storage forms of food e.g. plants store food in form of **starch** while animals store food in form of **glycogen**.

3. They are components of structures that provide mechanical support in organisms e.g. cellulose in cell walls, chitin in exoskeleton and lignin in xylem vessels and tracheids.

Classes of carbohydrates.

- ✓ They include:
 - A. Monosaccharides.
 - B. Disaccharides.
 - C. Polysaccharides.

A. MONOSACCHARIDES.

- ✓ They are simple sugars whose general formula is $(\text{CH}_2\text{O})_n$ where $n=6$.
- ✓ Therefore the chemical formula of a monosaccharide is $\text{C}_6\text{H}_{12}\text{O}_6$.
- ✓ Examples include:
 - i. Glucose.
 - ii. Fructose in ripe fruits and honey.
 - iii. Galactose in milk.

Functions /uses of monosaccharides.

- i. Used in respiration to provide energy.
- ii. When condensed they are storage forms of food e.g. plants store food in form of **starch** while animals store food in form of **glycogen**.

Properties of monosaccharides.

- i. They easily dissolve in water.
- ii. They are **reducing sugars** (when mixed with Benedict's solution they reduce copper (II) ions in the solution into copper (I) ions which are **brown in color**).
- iii. They have a sweet taste.
- iv. They form crystals/ they are crystallisable.

B. DISACCHARIDES.

- They are formed by combining two monosaccharides in the process called **condensation** releasing water.

- Examples include:

- i. Maltose

- ii. Sucrose.

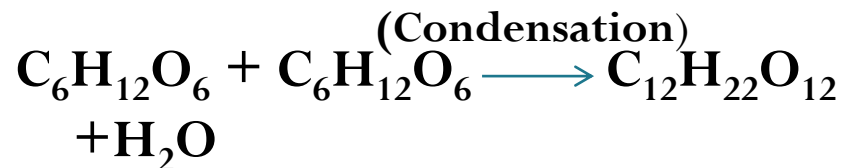
- iii. Lactose.

- i. **Maltose-** It is found in germinating seed and formed by combining two glucose molecules

Glucose + Glucose (Condensation)
Maltose + Water →

- ✓ Maltose is a reducing sugar.

Monosaccharide + monosaccharide →
Disaccharide + water



ii. **Sucrose**- it is found in sugarcane juice formed by combining **glucose and fructose**.

✓ Sucrose is non-reducing sugar.

(Condensation)

Glucose + Fructose $\longrightarrow$ Sucrose + Water

iii. **Lactose**- It is found in milk and formed by combining galactose and glucose.

✓ Lactose is a reducing sugar.

(Condensation)

Glucose + Galactose $\longrightarrow$ Lactose + Water

Characteristics/ properties of disaccharides.

- i. They are soluble in water.
- ii. They have a sweet taste/ form sweet tasting solutions.
- iii. Some disaccharides are reducing sugars e.g. maltose and lactose while sucrose is non-reducing sugar.
- iv. They can easily be broken down into monosaccharides through **hydrolysis**.

Hydrolysis of disaccharides.

- ✓ It involves disaccharides into monosaccharides/ simple sugars in the presence of water.
- ✓ Hydrolysis is brought about by:
 - i) Enzymes (in nature/ naturally)
 - ii) Heating with the acid e.g. Hydrochloric acid (in the laboratory)

(Hydrolysis)
Disaccharide + water $\longrightarrow$ Monosaccharide + Monosaccharide.

Examples

(Hydrolysis)
i. Maltose + Water $\longrightarrow$ Glucose + Glucose.

(Hydrolysis)
ii. Lactose + Water $\longrightarrow$ Glucose + Galactose.

(Hydrolysis)
iii. Sucrose + Water $\longrightarrow$ Glucose + Fructose.

C. POLYSACCHARIDES.

- ✓ They are made up of many monosaccharides.
- Examples include:
 - a) **Starch-** stored in plant tissues. Plants with a lot of **starch** include maize, wheat, potato and rice.
 - b) **Glycogen-** stored in animal tissues. It is synthesized from excess **glucose**.
 - c) **Cellulose-** found at cell walls of plant cells giving them definite shape.
 - d) **Chitin-** it is found on exoskeleton of arthropods and cell wall of fungal hyphae.
 - e) **Lignin-** it is found in xylem vessels and tracheids and provide mechanical support.

Characteristics / properties of polysaccharides.

- i. Are insoluble in water.
- ii. They do not have a sweet taste.
- iii. They are non-reducing sugars.
- iv. They are hydrolyzed into monosaccharides (by heating with acids or by enzymes)

Study question

1. Name the carbohydrate:
 - a) Present in abundance in germinating seed- **Maltose.**
 - b) Stored in plant cells- **Starch.**
 - c) Found on plant cell walls- **Cellulose.**
 - d) Found in animal tissues/muscles- **Glycogen.**
 - e) Found in blood- **Glucose.**

PRACTICAL 1.

Aim: Testing for starch.

Requirements:

1. Food substance in solution form.
2. Test tubes.
3. 10 ml measuring cylinder
4. Dropper
5. Iodine solution (Reagent)

Procedure.

1. Into a clean test tube, add 2ml of food substance.
2. Add 3 drops of iodine solution to the food substance and shake.

Observation.

- ✓ The colour turns/ changes (from brown) to **blue-black/ dark blue/ black.**

Conclusion.

- ✓ Starch present.

Note: If brown colour (of iodine) persists / is retained then starch is absent

PRACTICAL ACTIVITY 2.

Aim: To test for the presence of reducing sugar.

- Reducing sugars include glucose, fructose, galactose, maltose and lactose.

Requirements:

1. Food substance in solution form.
2. Benedict's solution (reagent).
3. Test tube.
4. Means of heating/ hot water bath.
5. Test tube holder.
6. 10 ml measuring cylinder.
7. Dropper.

Procedure:

1. Put 2 ml/cm³ of reducing sugar in a test tube.
2. Add equal amount of Benedict's solution.
3. Heat to boil.

Observations

- The colour changes from **blue to green to yellow and finally orange/ brown.**

Conclusion.

- Reducing sugar present.

N/B.

- If the colour changes to:

- i. **Green with no further change-** very little amount of reducing sugar is present.
 - ii. **Yellow** –average amount of reducing sugar present.
 - iii. **Orange/ brown-** high amount of reducing sugar present.
- If the **blue color (of Benedict's solution)** is retained, then reducing sugar is absent.
 - Reducing sugar changes copper sulphate in Benedict's solution to copper oxide (which is orange).

PRACTICAL ACTIVITY 3.

Aim: Testing for the presence of non-reducing sugar (e.g. sucrose).

Requirements:

1. Food substance.
2. Benedict's solution.
3. Dilute hydrochloric acid.
4. Sodium hydrogen carbonate.
5. Means of heating/ bunsen burner.
6. Dropper.
7. 10 ml measuring cylinder.

Procedure.

1. Put 2ml of the food substance into a clean test tube.
2. Add 4 drops of dilute hydrochloric acid and shake.
3. Boil the mixture.

4. Cool the mixture in cold water.
5. Add sodium hydrogen carbonate drop wise until fizzing stops.
6. Add equal amount of Benedict's solution to the mixture.
7. Heat the mixture to boil/ boil the mixture.

Observations

- The colour changes from **blue to green to yellow and finally orange/ brown.**

Conclusion.

- Non- reducing sugar present.

Points to note.

- Dilute hydrochloric acid is used/ added to hydrolyze non-reducing sugar to reducing sugar.
- Sodium hydrogen carbonate is used/ added to neutralize the acid.

2. PROTEINS.

- They contain carbon, hydrogen, oxygen and nitrogen (hence they are called **nitrogenous compounds**).
- Proteins may also contain other compounds e.g. phosphorus, sulphur and iron.
- They are made up of **amino acids** as building blocks i.e. amino acids form proteins.
- A protein consists of several amino acids joined together by a bond called **peptide bond**.
- Two amino acids combine to form **dipeptide molecule** in the process of **condensation**.
- When 2 amino acids combine they form a **dipeptide** joined by a **peptide bond**.

Structure of amino acid.

- It consists of
 - i. Amino group (NH_3)- consisting of hydrogen and nitrogen.
 - ii. Carboxyl group (COOH)- consisting of carbon, oxygen and hydrogen.
- A molecule consisting of few amino acids are called **peptides**.
- When peptides combine they form a **polypeptide** which makes up a protein.

- Therefore a protein is made up of one polypeptide or many polypeptides.
- Joining of amino acids to form peptides is called **condensation**.
- Break down of peptides to form amino acids is called **hydrolysis**.
- There are 20 naturally occurring amino acids which can be synthesized by plants.
- Human beings can only synthesize 11 amino acids in their bodies while 9 are supplied from diet.
- The amino acids that can be synthesized in the human bodies are called **non-essential amino acids**.
- Those amino acids that cannot be synthesized but are supplied from diet are called **essential amino acids**.
- Proteins that contain all the essential amino acids are called **1st class proteins e.g. animal proteins**.
- Proteins that lack one or more of the essential amino acids are called **2nd class proteins e.g. plant proteins**.

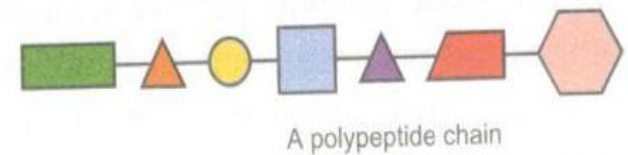
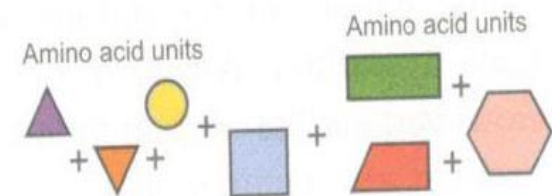
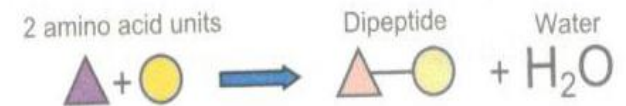
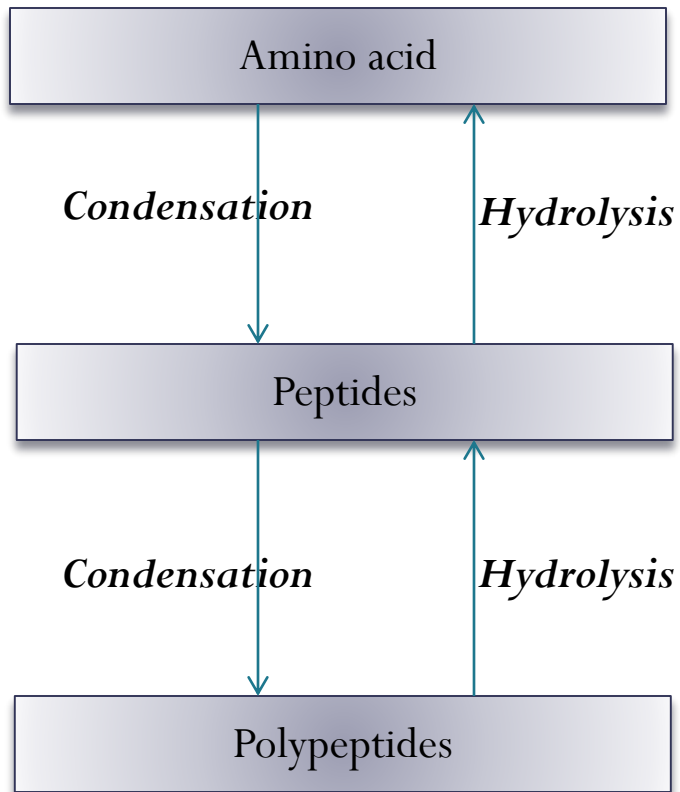


Fig 5.12: Diagrammatic representation of polypeptide formations

Properties of proteins.

1. They dissolve in water forming colloidal suspensions.
2. They are **denatured** by high temperatures above 40 degrees Celsius and extreme pH values. Denaturing changes the structure of protein molecules hence changing physical and chemical properties of protein.
3. Have acidic and basic properties hence described as **amphoteric**.
- ✓ They therefore react with acids and bases enabling them to form **conjugated proteins (i.e. proteins containing non-protein components)**.
- ✓ Examples of conjugated proteins include:
 - i) **Mucus** which contains a **carbohydrate**.
 - ii) **Haemoglobin** which contains **iron**.

Functions of proteins.

1. They are components of structures in living organisms (e.g. plasma/ cell membranes, connective tissue, hair, hooves, nails, muscle fibre, skeletal materials).
2. They are used for making, repair and replacement of worn out tissues in plants and animals.
3. They act as metabolic regulators (e.g. **enzymes** which speed up metabolic reactions, **hormones** which regulate body processes like growth, reproduction, **antibodies** that provide immunity against diseases).
4. They are broken down to give energy during starvation.

PRACTICAL ACTIVITY.

Aim: To test for proteins (biuret test).

Requirements:

1. Food substance in solution form
2. 1% Copper (II) sulphate solution.
3. 10% sodium hydroxide solution
4. Test tube
5. Droppers
6. 10 ml measuring cylinder.

Procedure:

1. Put 2ml of food substance into a test tube.
2. Add equal amount of 10% sodium hydroxide and shake.
3. Into the mixture, add 1% copper (II) sulphate dropwise and shake after every addition.

Observation:

- Colour changes to **purple**.

Conclusion:

- Proteins present.

Note: Sodium hydroxide is used to break the peptide bond.

If blue color is retained / persists then protein is absent.

3. LIPIDS (fats and oils).

- ✓ Fats are found in animals while oils are found in plants.
- ✓ Fats are solid at room temperature while oils are liquid at room temperature.
- ✓ They are like carbohydrates only that they have fewer number of oxygen molecules than carbohydrates.
- ✓ The building blocks of lipids are **fatty acids and glycerol** joined by **ester bond** in the process called **condensation**.
- ✓ The lipids can be broken down to form glycerol and fatty acids through **hydrolysis**.
- ✓ Fatty acids are of different types because they contain different fatty acids e.g. Oils are different in different plants because they have different fatty acids.

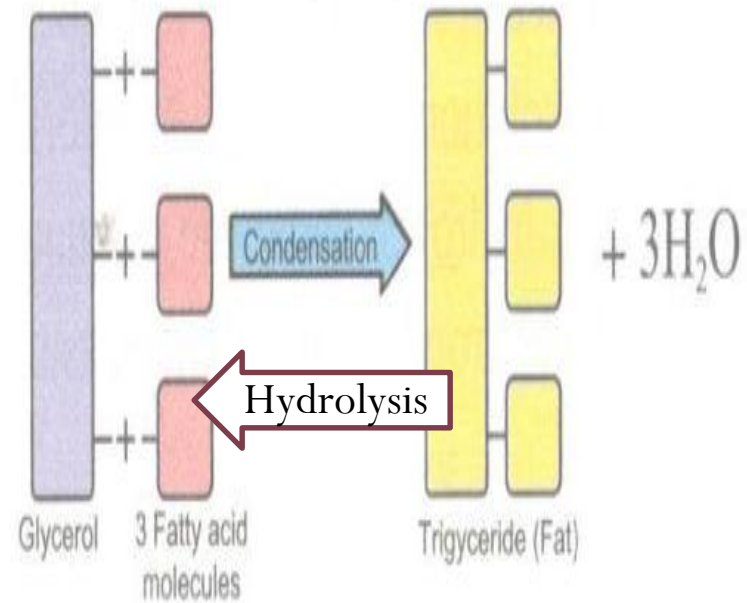


Fig 5.11: Diagrammatic representation of lipid formations.

Properties of lipids/ fats and oils.

1. Fats readily/ easily change into liquid when heated and oils solidify when subjected to low temperature.
2. They are insoluble in water.
3. They dissolve in organic solvents forming emulsions and suspensions.
4. They are inert (hence can be stored in tissues of organisms).

Functions of lipids/ fats and oils.

1. They are broken down/ oxidized to release/ give energy.
2. They are storage forms of food in the body of living organisms e.g. excess can be converted into fats for storage.

3. They are broken down/ oxidized to release metabolic water (to supplement water requirements in the body).
4. Lipids are components of plasma membrane and protoplasm while oils are storage structures in some seeds (e.g. ground nuts, castor seed, maize grains etc).
5. Fats are deposited under the skin as adipose tissue which insulates the body against heat loss/ prevents heat loss.
6. Thick adipose tissue in some aquatic organisms (e.g. whale and hippopotamus) makes them buoyant in water.
7. Fats are deposited around body organs (e.g. kidney, heart and back of eyeball) where it acts as shock absorber.
8. Lipid (wax) found on the cuticles reduce excessive water loss in plants.

PRACTICAL ACTIVITY.

Aim: Testing for the lipids (fats and oils).

A. The grease spot test.

Requirements:

1. Food substance.
2. Filter paper.
3. Bunsen burner.

Procedure:

1. Rub a little amount of food substance on a filter paper.
2. Hold the paper above the flame to dry taking care not to burn it.
3. Hold the paper against light.

Observation:

- A permanent translucent spot is formed.

Conclusion:

- Lipids (fats and oils) present.

N/B If the permanent translucent spot is absent, then lipids are absent.

B. Emulsion test.

Requirements:

1. Food substance.
2. Ethanol/alcohol.
3. 2 test tubes.
4. 10ml measuring cylinder.

Procedure:

1. Into a clean test tube, put a little amount of food substance.
2. Add 2 ml of ethanol/alcohol to 2ml of the food substance and shake thoroughly.
3. Transfer the contents of the test tube into another test tube half filled with water.

Observation:

- Formation of a white emulsion.

Conclusion:

- Fats or oils present.

Note:

- Emulsification of fats occurs in the duodenum to increase the surface area for digestion by enzymes.

4. ENZYMES.

- They are organic **biocatalysts** i.e. they speed up or slow down the rate of chemical reactions but they are not used up in the process.
- Enzymes are protein in nature and are produced in living cells.

Types of enzymes

- a) **Intracellular enzymes-** they are secreted/ produced and used within the cells that produce them e.g. respiratory enzymes.
- b) **Extracellular enzymes-** they are secreted/ produced by cells but used outside the cells that produce them e.g. digestive enzymes.

Naming of enzymes

- a) **Trivial naming-** naming of enzymes based on the persons who discovered them. The names of such enzymes end with suffix- **in**, e.g. **Pepsin, Rennin, Trypsin, Ptyalin.**

- b) **Use of suffix- ase-** the suffix –ase is added to the type of food/ substrate or reaction the enzymes catalyze e.g.

Food/ substrate

Enzyme

Carbohydrate	Carbohydrase
Amylose/ starch	Amylase
Sucrose	Sucrase
Maltose	Maltase
Protein	Protease
Lipids	Lipase

Chemical reaction

Enzyme

Hydrolysis	Hydrolase
Oxidation	Oxidase
Reduction	Reductase

Properties of enzymes.

1. They are affected by changes in temperature and pH (because they are protein in nature).
2. They are substrate specific i.e. they act on specific substrate.
3. They are very efficient hence required in small quantities.
4. They are not affected by reactions they catalyze hence they are available for reuse.
5. Most reactions catalyzed by enzymes are reversible.

Importance of enzymes.

- They control and regulate biochemical reactions in the body so that they proceed at a pace suitable for sustaining life.
- This is because the biochemical reactions in body cells are too fast and others too slow. This ensures order in living systems.

N/B

- The speed of enzyme-catalyzed reaction is called **enzymes turnover**.

FACTORS AFFECTING ENZYME CONTROLLED REACTIONS/ ENZYME ACTIVITY.

1. **TEMPERATURE**- enzymes work best within narrow range of temperature (between 35°C- 40°C).
- Increase in temperature increases the rate of enzyme activity up to optimum point. Optimum temperature gives maximum enzyme activity.
- Above the optimum temperature the rate of enzyme activity decreases sharply because **higher temperature denatures / destroys the enzymes** making them non effective.
- When the temperature decrease, the rate of enzyme activity decreases **because enzymes are inactivated.**
- Low temperature does not destroy the enzymes because when temperature is increased again the enzymes become active.

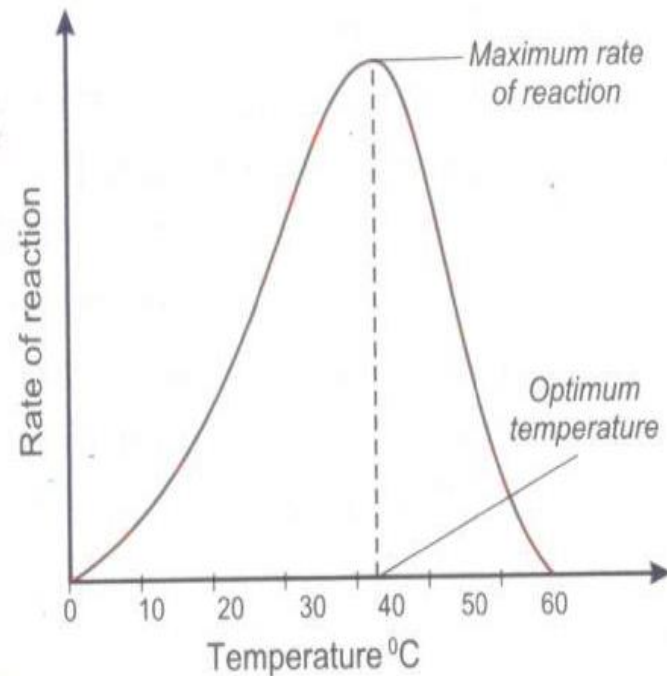


Fig.5.13: Effect of temperature on the rate of enzyme activity

2. **pH-** This is acidity or alkalinity of an substance.

- Most enzymes work best at optimum pH of 7 but others work best in acidic conditions others in basic conditions.
- Change of pH from optimum decreases the rate of enzyme activity.
- Extreme changes of pH range from the optimum denatures the enzymes hence the rate of enzyme activity stops.

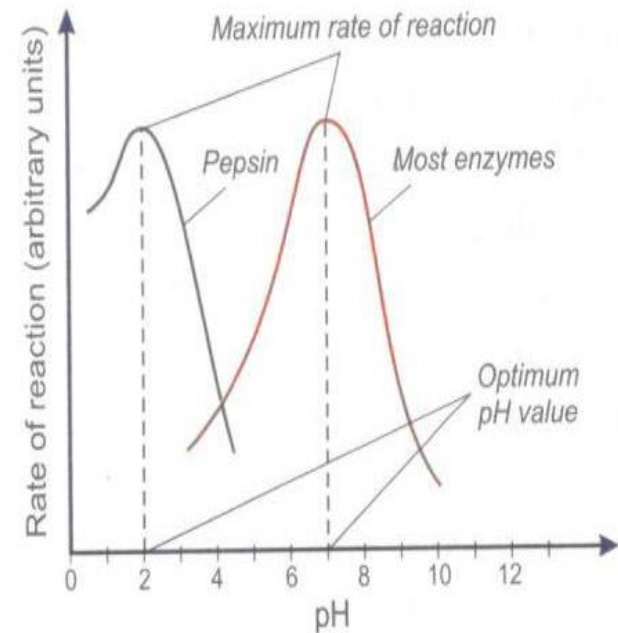


Fig 5.14: Effect of pH on the rate of an enzyme controlled reaction.

3. SUBSTRATE CONCENTRATION AND ENZYME CONCENTRATION-

when the substrate concentration is increased, the rate of enzymatic reaction also increases up to a certain/ maximum level.

- Further increase in concentration of the substrate does not increase the rate of enzymatic reaction.
- This is because **all active sites of an enzyme are occupied by the substrate**. At this point, the enzyme concentration becomes a limiting factor.
- An increase in amount or concentration of enzyme molecules increases the rate of enzyme reaction. This is because of an **increase in the number** of active sites of enzymes.

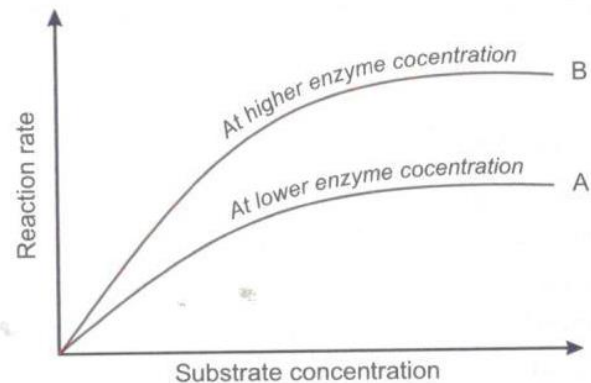


Fig 5.15: Effect of substrate and enzyme concentration on the rate of an enzyme controlled reaction.

Graph A shows the effect of increasing substrate concentration on the rate of an enzyme controlled reaction.

Graph B shows effect of increased enzyme concentration on the rate of an enzyme controlled reaction.

4. **ENZYME INHIBITORS-** they are chemical substances that prevent an enzyme from catalyzing a reaction hence decrease the rate of enzyme activity.
- This is because **they compete for active sites** of enzymes.
 - There are two types of inhibitors namely:
 - ✓ Competitive inhibitors.
 - ✓ Non-competitive inhibitors.
 - a) **Competitive inhibitors-** They have the same shape as that of the substrate and compete for the same active sites of the enzyme. This slows down the rate of enzyme activity.
 - The inhibitor stays attached to the active sites of the enzyme preventing the enzyme substrate from binding onto the active sites, hence slowing down the rate of reaction.
- The competitive inhibition can be overcome by:
 - i. Increasing the substrate concentration.
 - ii. Reducing the amount of inhibitor.
 - b) **Non-competitive inhibitors-** they attach themselves onto the enzyme molecules changing the shape of the active sites of the enzymes.
 - The substrate molecules are not able to bind onto the active sites of the enzymes.
 - They do not compete for the active sites of the enzymes with the substrate molecules.
 - Examples of non-competitive inhibitors include **cyanide, mercury, silver-arsenic compounds.**

5. **ENZYME CO-FACTORS-** are non-protein substances that activate the enzymes hence increase the rate of enzyme activity.

- Examples include **metallic ions e.g. magnesium ions, zinc ions, copper, calcium ions, chloride ions, molybdenum ions, manganese ions**

- They are required in small quantities.

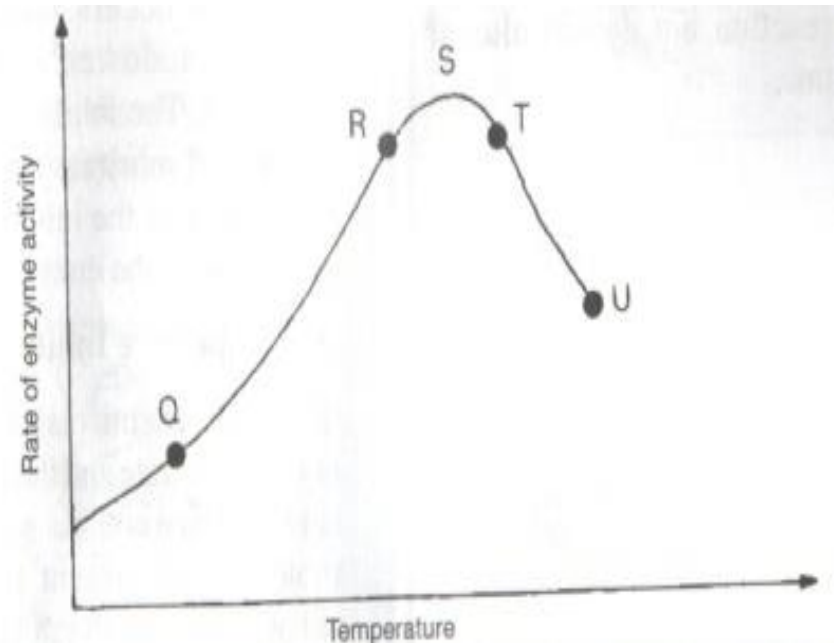
6. **ENZYMES CO-ENZYMES-** are organic co-factors that activate enzymes hence increase the rate of enzyme activity **e.g. vitamins.**

7. **PRODUCT CONCENTRATION-** low product concentration increases the rate of enzyme activity while high product concentration reduces the rate of enzyme activity.

8. **ENZYME SUBSTRATE SPECIFICITY-** enzyme act on specific substrate.

Study question

- The figure below shows the rate of enzyme action in relation to changes in temperature. Study it and answer the questions that follow.



1. Explain giving reasons, the rate of enzyme action :

i. Between Q and R.

✓ **The rate of enzyme activity increases with increase in temperature, because enzymes are activated.**

ii. At S.

✓ **At S there is maximum rate of enzyme reaction because it is the optimum temperature.**

iii. Between T and U.

✓ **There is drastic drop in the rate of reaction of enzymes because very high temperature denatures /destroys the enzymes.**

2. Other the factor being investigated above, state three other factors that affect the rate of enzyme activity.

✓ pH

✓ Substrate concentration.

✓ Enzyme concentration.

✓ Enzyme co-factors.

✓ Enzyme co-enzymes

✓ Enzyme inhibitors.

✓ Product concentration.

✓ Enzyme specificity

3. State the collective name of the enzymes that work on:

a) Carbohydrates- **carbohydrases.**

b) Proteins- **proteases**

c) Lipids- **lipases**

NUTRITION IN ANIMALS.

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- Nutrition in animals is called **heterotrophism**.

Heterotrophism/ heterotrophic mode of nutrition.

- ✓ This is a type of nutrition in which organisms feed on other organisms or feed on already manufactured food.
- ✓ These type of organisms are called **heterotrophs**.

Types of heterotrophs.

They include:

- i. **Herbivores**- they feed on plant materials e.g. cows, goats, sheep, grasshoppers.

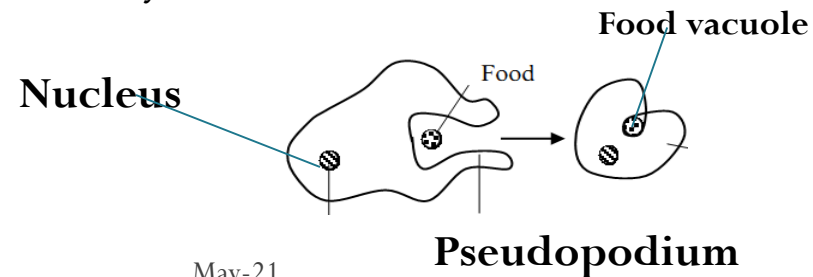
- ii. **Carnivores**- they feed on flesh/ meat/ other animals e.g. lions, dogs, hyenas, eagles.
- iii. **Omnivores**- they feed on both plants and animals/ flesh e.g. man and pigs.
- iv. **Saprophytes**- they feed on dead decaying matter leading to decomposition e.g. fungi and bacteria.
- v. **Parasites**- they live on or in other organisms called hosts and depend on them for nutrients.

Types/modes of heterotrophism

- A. Holozoic nutrition.
- B. Saprophytic nutrition.
- C. Phagocytosis.
- D. Parasitic nutrition/ parasitism.
- E. Symbiosis

May-21

- a) **Holozoic nutrition-** this is a type of nutrition where organisms/ animals ingest/take in, digest and assimilate complex food materials. It is common in mammals and birds.
- b) **Saprophytism-** this is a type of nutrition where organisms obtain nutrients from dead decaying matter causing decomposition.
- ✓ They release enzymes that break down decaying matter into simpler soluble substances that are absorbed directly into the body. It is common in bacteria and fungi.
- c) **Parasitism-** this is a mode of nutrition where one organism (parasite) feeds on or obtains nutrients from the tissues of another living organism (the host) **e.g. ticks and roundworms.**
- d) **Symbiosis-** this is an association where two organisms live together and mutually benefit from each other **e.g. Rhizobium bacteria and leguminous plant.**
- e) **Phagocytosis-** This is a type of nutrition where single celled organisms **e.g. amoeba and some white blood cells** feed on solid food materials.
- ✓ They engulf food material by forming pseudopodia and enclose it in food vacuole.
 - ✓ The enclosed food is digested by enzymes into soluble substances that are absorbed into the cytoplasm.



Animal dentition and dental formula.

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- ✓ **Dentition** refers to the description of the types, arrangement and specialization of teeth in animals.
- ✓ **Dentition** is related to the type of food the animal feeds on.

Types of dentition.

- i. **Homodont dentition**-in this type, all teeth have similar shape and size e.g. in fish, frogs, crocodiles.
- ii. **Heterodont dentition**-in this type, teeth have different shape and size e.g. in mammals.

- ✓ **Dental formula** describes the number, type and position of teeth in the jaw of the animal.
- ✓ The number usually given is for half of each jaw.
- ✓ To get the total number of teeth in an animal, the total number in the formula is multiplied by 2.

Types of teeth.

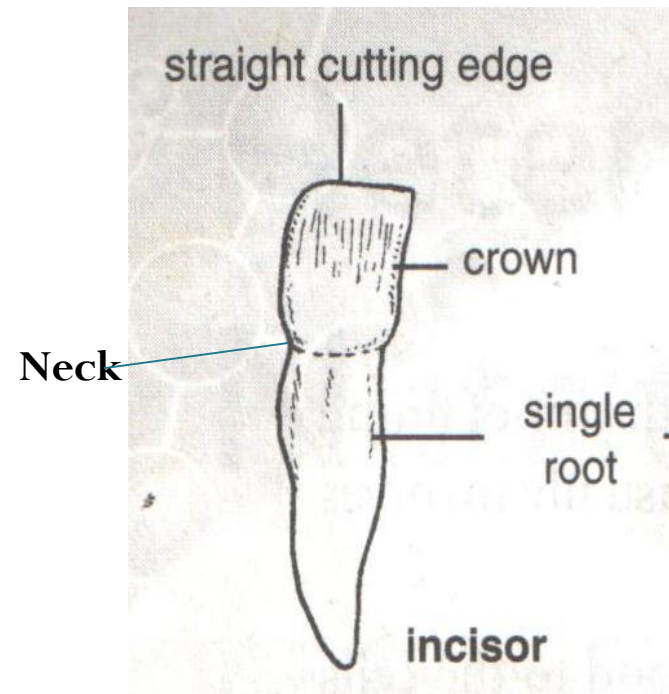
- ✓ There are four types of teeth namely:
 - A. Incisors.
 - B. Canines.
 - C. Premolars.
 - D. Molars.

A. INCISORS.

- ✓ Incisors are located at the front of the jaw.

Adaptations of incisor.

1. The crown is chisel-shaped/ flat and sharp for holding, biting and cutting food.
2. Have one root to support them in the jaw.

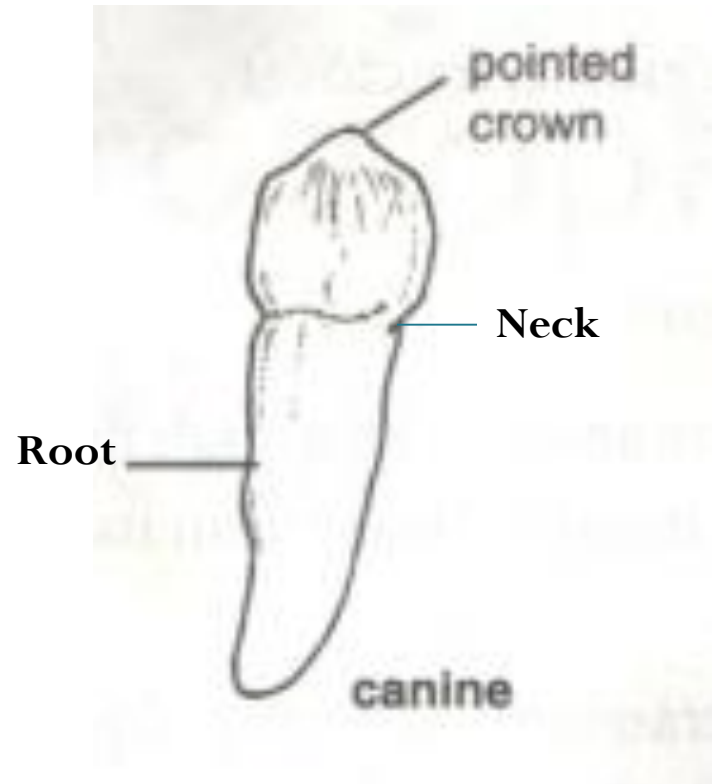


B. CANINES

- They are located next to the incisors.

Adaptations of canine.

1. It has curved and pointed crown for holding/ seizing prey and tearing flesh.
2. Have one pointed root for firm support in the jaw.

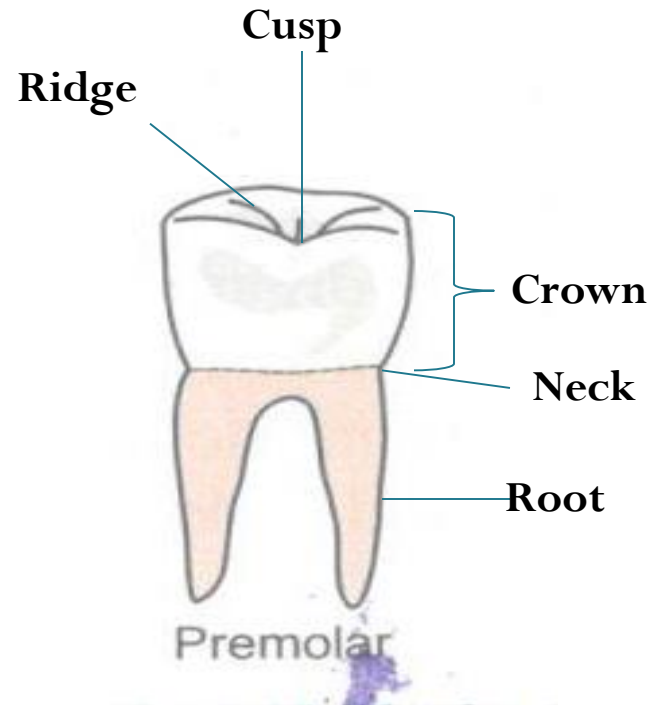


C. PREMOLARS.

- ✓ They are located after the canines towards the back of the jaw.

Adaptations of premolar.

1. The crown is **broad**, **ridged with cusps** to increase the surface area for crushing and grinding food.
2. It has 2 roots for firm support in the jaw.

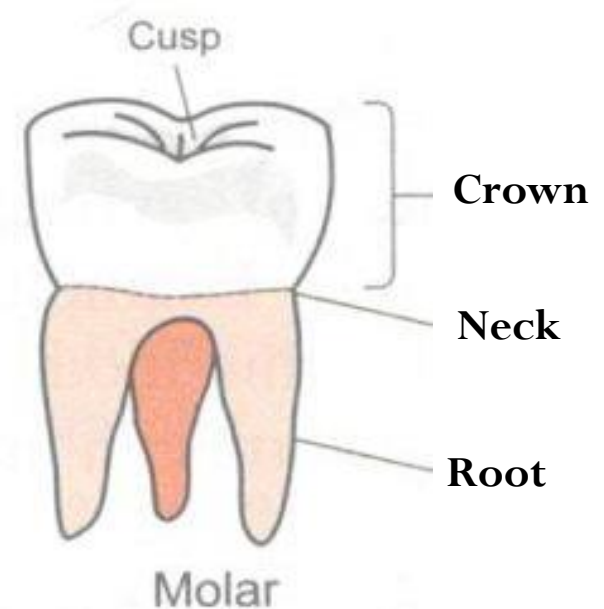


D. MOLARS

- They are located at the back of the jaw

Adaptations of premolar.

1. The crown is **broad**, **ridged with cusps** to increase the surface area for crushing and grinding food.
2. It has 3 roots for firm support in the jaw.



HERBIVORE DENTITION

✓⁷⁷ Herbivores are animals which feed on vegetation e.g. cow, goat, sheep, donkey, zebra e.t.c.

✓ The mode of nutrition is called **herbivorous**.

✓ The dental formula is $i \frac{0}{3} \quad c \frac{0}{1} \quad pm \frac{3}{2} \quad m \frac{3}{3} = 30$

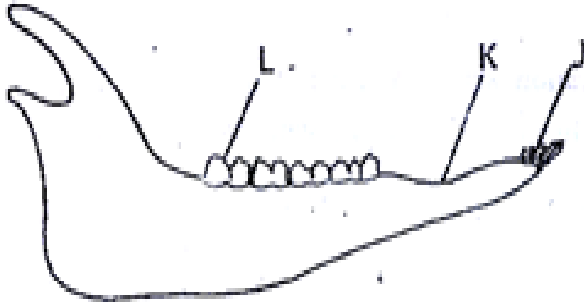
Adaptations of herbivore dentition.

1. They lack upper incisors and canines instead they have a rough **horny pad** against which grass/ vegetation is pressed and cut by lower incisors.
2. They have **diastema (a gap in the lower jaw between canines and premolars)** which provides a space for the tongue to manipulate grass/ vegetation/ plant materials (so as to separate the newly cut vegetation from that which is being chewed at the back of the mouth).

3. It lacks canines to facilitate side ways movement.
4. The lower incisors have chisel shaped crown with sharp edge for cutting vegetation.
5. The crown of premolars and molars has broad surface with cusps and ridges to increase surface area for grinding of food/ vegetation.
6. The teeth grow continuously throughout the life of the animal to replace worn out enamel worn out due to continuous grinding.
7. The jaws move side to side to enable premolars and molars to grind vegetation.

Study question.

- ✓ The diagram below represents the lower jaw of a mammal



- a) Name the mode of nutrition of the mammal whose jaw is shown. (1mk)
 - ✓ ***Herbivorous.***
- b) Give a reason for your answer in a) above. (1mk)
 - ✓ ***It has diastema.***
- c) Name and give the function of the toothless gap labeled K. (2mks)
 - ✓ ***Diastema- to provide room for manipulation of food to separate ground and unground food.***

- d) State one structural and one functional difference between the teeth labelled J and L. (2mks)

Structural

- ✓ ***Tooth J is narrow / sharp / chisel like while tooth L is broad / ridged.***
- ✓ ***J has one root while L has 3 roots.***

Functional

- ✓ ***Tooth J is used for cutting food while tooth L is used for crushing food;***
- e) Name the substance that is responsible for hardening of teeth. (1mk)
 - ✓ ***Calcium phosphate / calcium carbonate.***

OMNIVORE DENTITION

- ✓ Omnivores are animals which feed on both vegetation and flesh e.g. man.
- ✓ The mode of nutrition is called **omnivorous**.
- ✓ The dental formula is $i \frac{2}{2} \quad c \frac{1}{1} \quad pm \frac{2}{2} \quad m \frac{3}{3} = 32$

Adaptations of omnivore dentition.

1. The incisors are flat and chisel shaped for cutting and biting food.
2. The canines are small and pointed for tearing of food.
3. Premolars and molars have cusps on the upper surface of the crown to increase the surface area for grinding and crushing of food.

- ✓ There are two sets of teeth in humans:

- i. **Milk teeth**- they are first set of teeth and are 20 in number. They are lost between the age of 6-12 years.
- ii. **Permanent teeth**- they replace the milk teeth are 28 in number.
- iii. **Wisdom teeth**- they are back molars that appear last at the age of 17-25 years. They are 4 in number

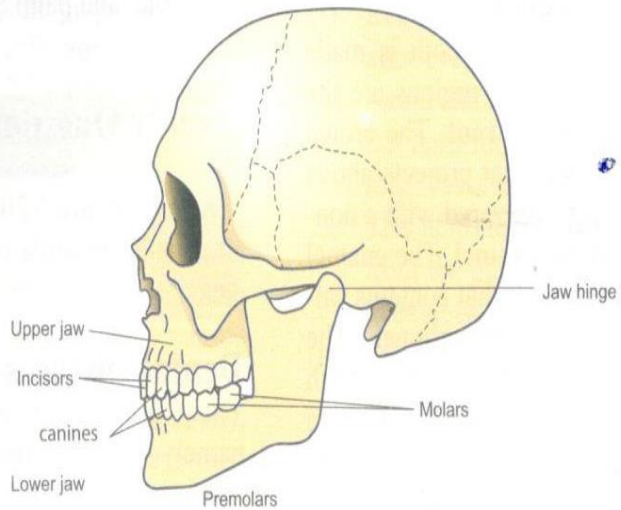


Fig 5.20 (a): A human skull and dentition.

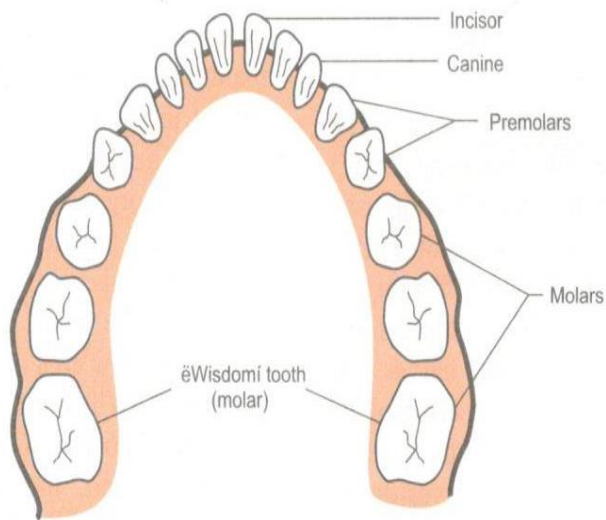
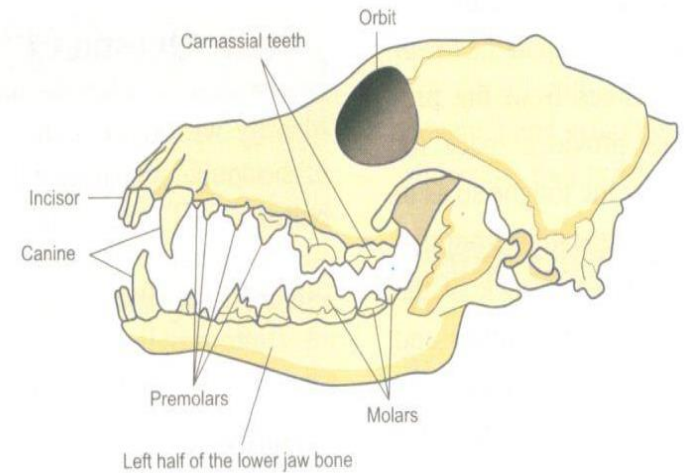
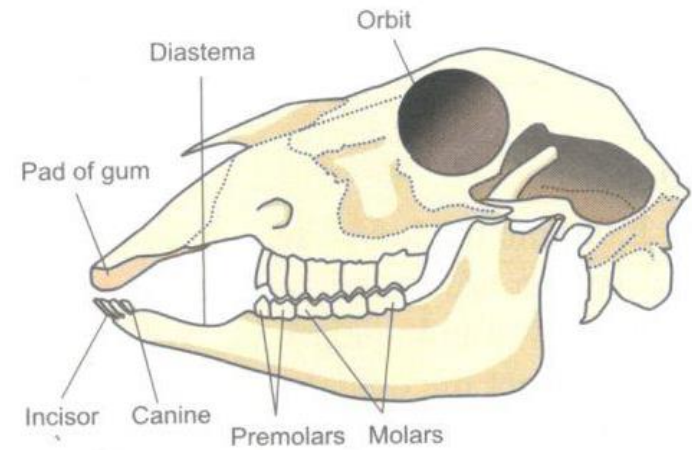


Fig 5.20 (b): Arrangement of different types of teeth in the human jaw.



CARNIVORE DENTITION

- ✓ Carnivores are animals which feed on flesh e.g. lion, dog, cheetah, cat e.t.c.
- ✓ The mode of nutrition is called **carnivorous**
- ✓ The dental formula is $i \frac{3}{3} \quad c \frac{1}{1} \quad pm \frac{4}{4} \quad m \frac{2}{3} = 42$.

Adaptations of carnivore dentition.

1. The incisors are chisel-shaped, small and closely fitted to seize/ hold the prey and stripping flesh from the bone.
2. The canines are long, curved and sharply pointed to hold, kill and tear the prey.
3. The crown of premolars and molars has broad surface with cusps and ridges to increase surface area for grinding flesh.
4. The **carnassial teeth** (modified last premolar in the upper jaw and first molar in the lower jaw) have smooth sides and sharp edges to slice through flesh and crush bones.
5. The jaws are attached to powerful muscles to crush the bones.

STRUCTURE OF A TOOTH

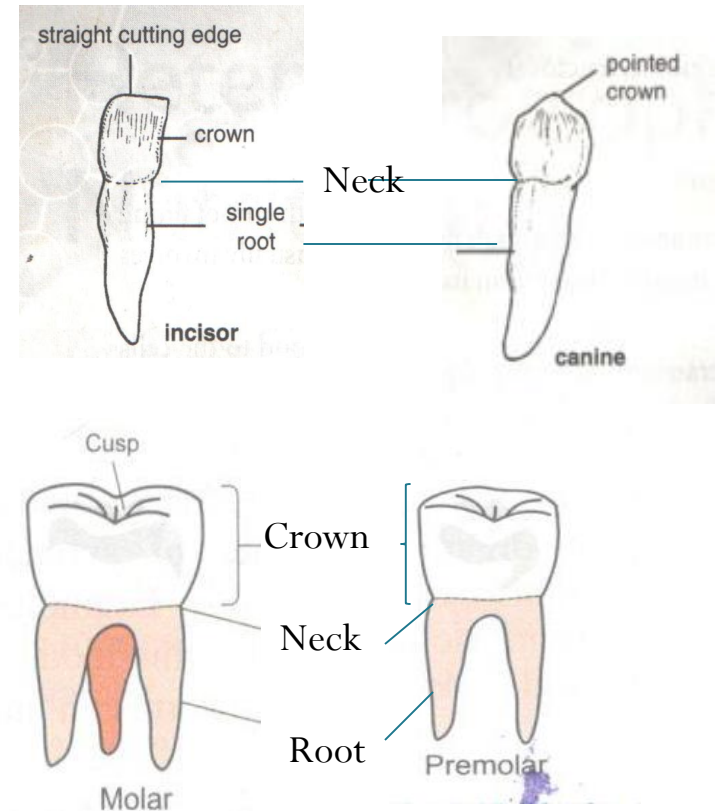
EXTERNAL STRUCTURE.

✓ It consists of:

1. **The crown-** the part that projects above the gum. It is covered by a hard and white non-living layer called **enamel**.
- ✓ The enamel is hardened by **calcium phosphate and calcium carbonate**.

Role/ functions of enamel.

- a) It protects internal structures from mechanical/ physical injury.
- b) It provides a hard surface for biting and grinding of food.
2. **The neck-** it is a parts between the crown and root. It is covered by gum.
3. **The root-** it is part of the tooth that is firmly fixed into the jaw by cement.



INTERNAL STRUCTURE.

1. **Enamel**- is hardened by **calcium phosphate and calcium carbonate.**

Role/ functions of enamel.

- i. It protects internal structures from mechanical/ physical injury.
- ii. It provides a hard surface for biting and grinding of food.

2. **Dentine**- it is located below the enamel and extends to the root.

Functions/ role of dentine.

- i. It is made up of living cells that give rise to the enamel.

3. **Pulp cavity**- it is located within the dentine.

Functions/ role of pulp cavity.

- i. It has blood vessels/capillaries to supply nutrients and oxygen to the cells of the tooth and remove waste products (nitrogenous wastes and carbon (IV) oxide).
- ii. Has sensory nerves/ nerve cells that detect heat, cold and pain.

4. **Cement**- It holds the tooth firmly into the jaw bone.

5. **Periodontal membrane**- It is found between the cement and jaw bone.

Functions/ role of periodontal membrane.

- i. Contains living cells that secrete the cement.
- ii. They act as shock absorbers i.e. allow some slight movement during chewing to avoid breakage.

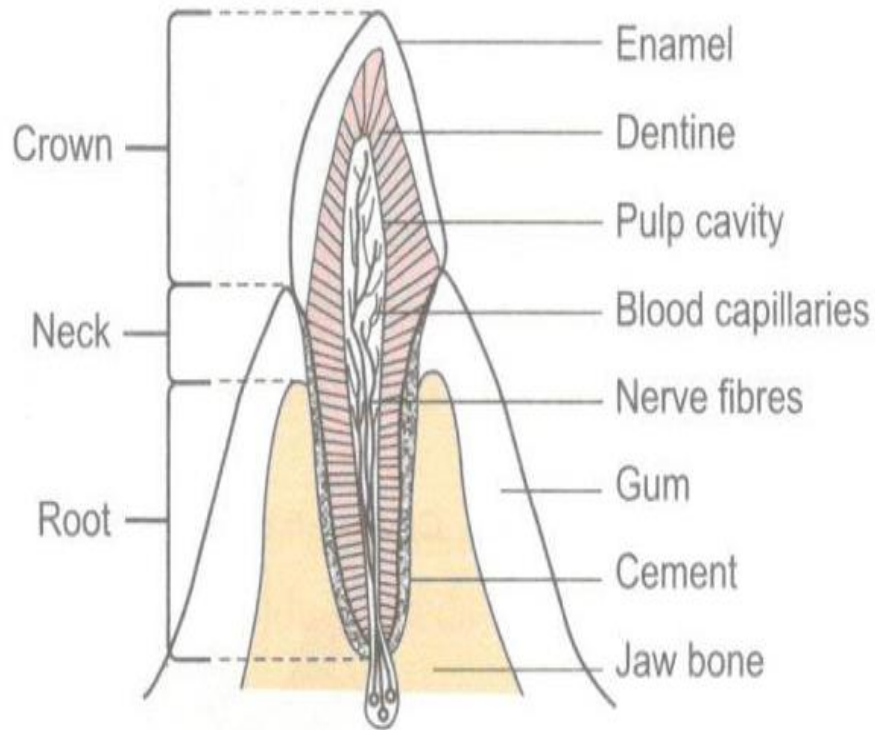
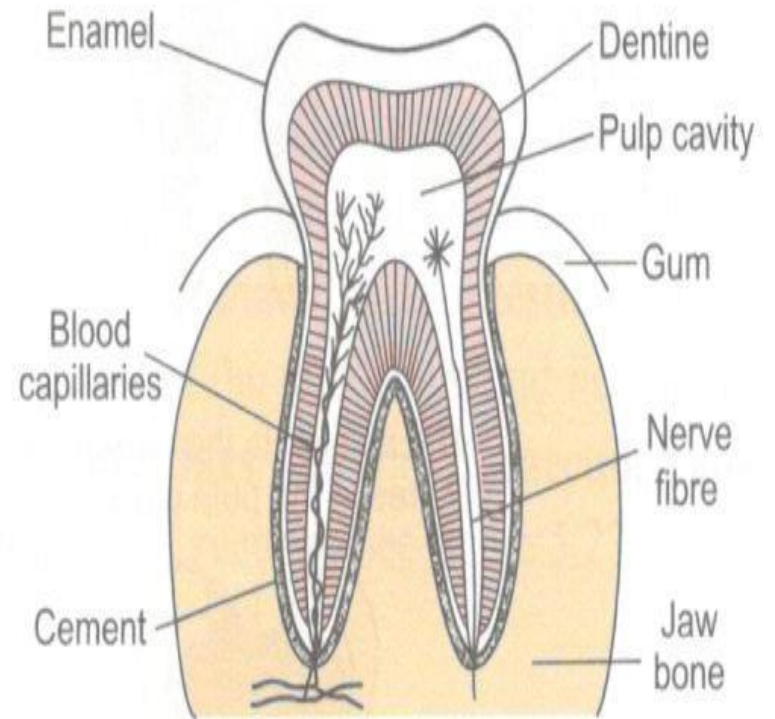


Fig 5.21: (a) Vertical section through an incisor tooth.



(b): Vertical section through a molar tooth.

DENTAL DISEASES

- ✓ Dental diseases include:
 - A. Dental carriers/ tooth decay.
 - B. Periodontal disease.
- A. **Dental carriers.**
 - ✓ It is caused by **plaque** (mixture of sugars, starch and micro-organisms) which accumulates between the tooth.
 - ✓ The micro-organisms breakdown sugars in the plaque to form/ produce acids.
- ✓ The acids react with the enamel and dentine causing them to dissolve forming a hollow cavity.
- ✓ When the decaying process continues to the pulp cavity, it affects the nerves leading to pain/tooth ache.
- ✓ In serious cases, the pulp cavity may be destroyed and the infection spread to the gums.

Prevention of dental carriers.

1. Avoiding too much sweet and sugary food.
2. Taking a diet rich in calcium and vitamin D.
3. Eating hard foods.
4. Cleaning the teeth regularly.
5. Filling of cavities to prevent further decay.
6. If the cavities affect the pulp cavity, root canal procedure can be performed by a dentist.
7. Extraction of the tooth by the dentist in serious cases.

Decay of enamel starting

Nerve fibres in the pulp cavity



Decay of enamel continues and affects the dentine



Decay affects the nerve fibres in the pulp cavity



Cavity left by the decay



Abscess (swelling) on the root



Fig 5.22: Progressive stages in tooth decay.

B. PERIODONTAL DISEASE

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- ✓ It affects the gum, caused by bacterial infection.
- ✓ The disease makes the gums to become soft and flabby, so that they do not support the teeth.

Types of periodontal disease.

1. **Gingivitis-** characterized by reddening of gums, bleeding of gums and presence of pus in the gums.
2. **Pyorrhoea-** is a condition where the teeth become loose due to infection and the teeth are finally lost.

Control of periodontal diseases.

1. Eating balanced diet and rich in vitamin A and C.
2. Brushing the teeth regularly (to encourage blood circulation)
3. Dental hygiene

Healthy practices that minimize dental diseases.

1. Regular brushing/ cleaning of teeth after every meal.
2. Avoid too much sugary foods.
3. Eating hard foods e.g. raw carrots, cassava, yams and sugarcane. This helps to exercise the teeth and remove soft materials from gums and teeth.
4. Taking a diet rich in calcium, phosphate, and vitamins A, C and D.
5. Teeth should be used for proper purpose. They should not be used to open beverage bottles or crack hard nuts.
6. Regular dental check up.

DIGESTIVE SYSTEM IN MAN.

✓ It is used for **digestion** and **absorption** and consists of:

A. Alimentary canal/ gut/ digestive tract.

B. Associated organs

✓ The alimentary canal/ gut/ digestive tract consists of:

i. Mouth.

ii. Oesophagus/ gullet.

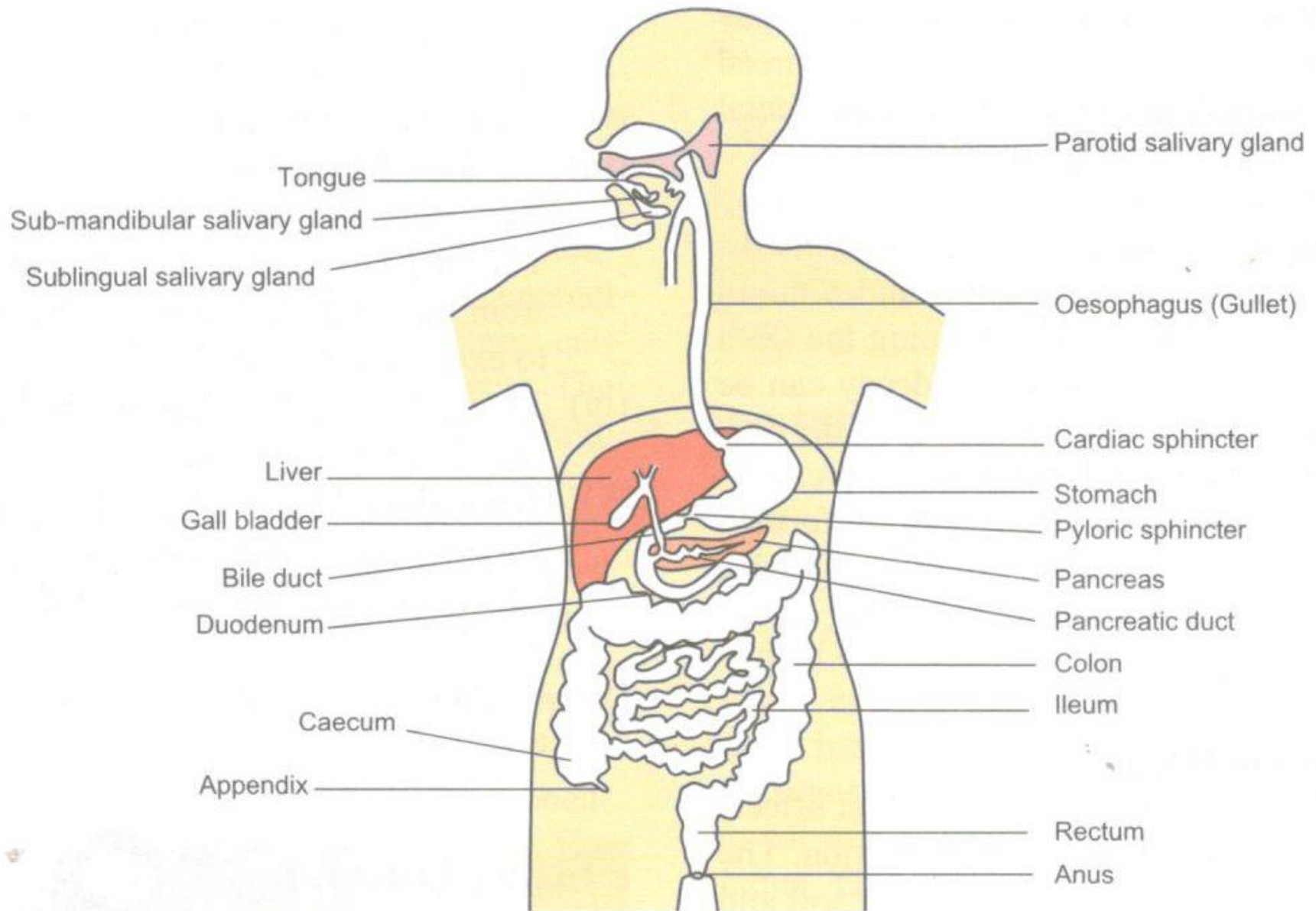
iii. Stomach.

iii. Small intestines (duodenum and ileum).

iv. Large intestines/ colon.

v. Rectum.

■ Along its length there are associated organs e.g. liver, gall bladder, pancreas, digestive glands.



DIGESTION

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- ✓ Digestion refers to mechanical/ physical and chemical breakdown of complex food material into simpler forms that can easily be absorbed into the body.
- ✓ Physical/ mechanical digestion is done by teeth in the **mouth** and by bile salts in the **duodenum**.
- ✓ Chemical digestion is done by the enzymes.
- ✓ When digestion occurs inside the cells it is called **intracellular digestion** e.g. in amoeba and white blood cells.
- ✓ When digestion occurs outside the cells where enzymes are secreted onto food is called **extracellular digestion** e.g. in man.

A. DIGESTION IN THE MOUTH.

- ✓ The ingested food is chewed by teeth into simpler particles. Chewing and grinding of food is called **mastication**.
- ✓ Chewing increases the surface area for action of enzymes on food. The tongue rolls and mixes food with saliva.
- ✓ There are three pairs of salivary glands, namely:
 1. **Sub-mandibular glands**- located near the back of the lower jaw which produce enzyme *Amylase*.
 2. **Sublingual glands** – located below the tongue which produce mucus.
 3. **2 parotid glands**- located on each side of the mouth and below the ear/ on the cheeks which produce enzyme *Amylase*.

✓ Saliva contains the following:

1. **Mucus/mucin-** which lubricates food.
 2. **Water-** which softens food, acts as a solvent/ dissolves food and moistens food/mouth.
 3. **Enzyme amylase/ptyalin-** which converts starch into maltose/digests starch.
- Saliva is slightly alkaline to provide optimum action of salivary amylase.

✓ Therefore the following are the functions of saliva:

1. Lubricates food.
2. Digestion of starch.
3. Moistens food/mouth.
4. Softens food.
5. Provides alkaline medium for action of enzyme (salivary amylase).
6. Dissolves food.

95 ✓ The tongue mixes food with saliva and rolls food into boluses pushes them to the back of the mouth/ pharynx for swallowing into the stomach.

✓ During swallowing the soft palate is raised to open the gullet and close nasal cavity while the **epiglottis** closes preventing food from entering into the **trachea/ wind pipe**.

✓ Food moves down the gullet/ oesophagus through the process called **peristalsis**.

✓ **Peristalsis** refers to involuntary movement of food in alimentary canal/ gut.

✓ It is brought about by contraction and relaxation of circular and longitudinal (smooth) muscles of the gut/ alimentary canal.

Adaptation of oesophagus/ gullet.

1. It is made up of smooth muscles with contract and relax to facilitate peristalsis.
2. Inner lining has goblet cells that secrete mucus to lubricate food hence facilitate smooth movement of food/ boluses.

Study question

- *How is the mouth adapted/ suited to its function?*
- ✓ Has teeth for chewing / grinding food to increase surface area for digestion by enzymes.
- ✓ Has salivary glands which secrete saliva which lubricate, soften, moisten, dissolve food and provide alkaline medium for action of enzymes.
- ✓ Saliva contains salivary amylase/ ptyalin which digest starch into maltose.
- ✓ It has muscular tongue to mix food with saliva and roll food into boluses and pushes them to the back of the mouth (for swallowing).

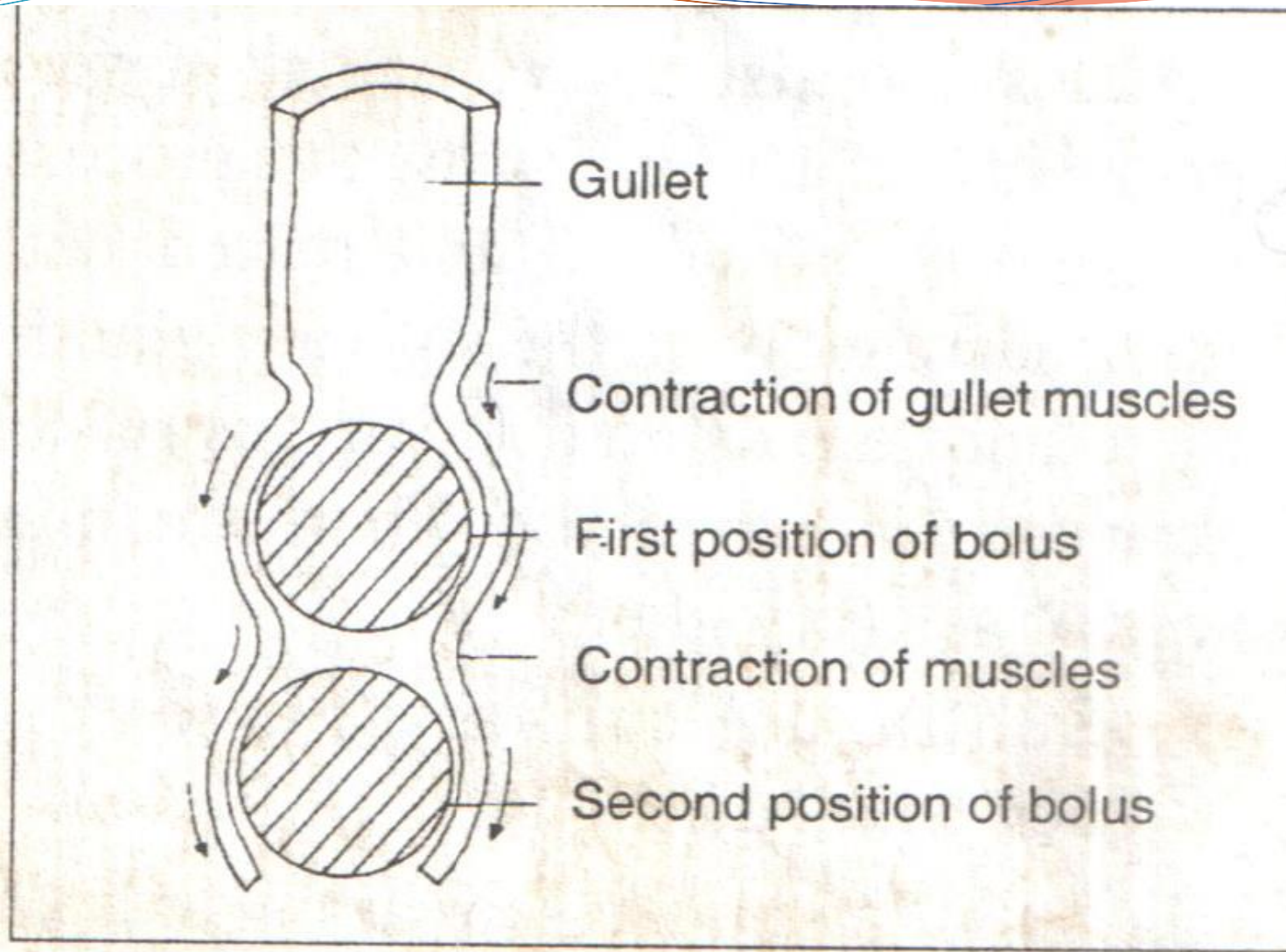


Fig. 5.25: Diagrammatic representation of peristalsis

B. DIGESTION IN THE STOMACH.

- ✓ Food from the oesophagus enters the stomach through the **cardiac sphincter muscle** which closes to prevent the food from moving up the oesophagus.

Role of stomach.

1. Churning of food.
2. Digestion of food/ proteins.
3. Absorption of alcohol, some water, water soluble vitamins (Vitamin B and C) and water soluble salts.

- ✓ The stomach walls have thick muscles (circular and longitudinal) which contract and relax to mix food into **chyme** in the process called **churning**.
- ✓ Churning helps to mix food with digestive enzymes.
- ✓ The presence of food in the stomach/ smell/ taste of food makes the stomach to secrete **gastrin hormone**.
- ✓ This hormone stimulates the **gastric glands** on the stomach walls to secrete **gastric juice**.

- ✓ In the stomach food is mixed with gastric juice which contains:
 1. **Pepsin enzyme**- It catalyzes/ speeds up conversion of proteins into peptides/ peptones.
- ✓ It is secreted by **peptic cells** in inactive/ precursor form called **pepsinogen** to avoid digesting / auto digestion of the stomach walls/lining (when no food is present in the stomach).
- ✓ **Pepsinogen** is converted into **pepsin** by **hydrochloric acid** present in the stomach.

Study question.

How is the stomach adapted to:

- i. **Churning**- The stomach walls have thick muscles which contract and relax.
 - ii. **Protein digestion**- The stomach walls contain gastric glands which secrete gastric juice containing Pepsin and Rennin.
2. **Rennin enzyme**- It is secreted in inactive/ Precursor form called **prorennin** by gastric glands in young mammals.
- ✓ Rennin which is abundant in young children converts soluble milk protein (caseinogen) into insoluble form (casein)/ curdles milk.
 - ✓ Curdling of milk provides enough time for digestion.

3. **Mucus-** secreted by goblet cells to protect the stomach walls against corrosion by hydrochloric acid and auto digestion by enzymes.
4. **Dilute hydrochloric acid-**
It is produced by cells of the stomach walls.
 - i. It provides acidic medium suitable for action of pepsin and rennin enzymes.
 - ii. It kills any bacteria which may be present in food.
 - iii. It converts inactive forms of rennin and pepsin into active forms.

Adaptation of stomach

1. It has muscular wall which contract and relax to facilitate churning of food into chyme.
2. The stomach lining has goblet cells that secrete mucus to protect them against auto digestion by enzymes.
3. The inner lining has gastric glands that secrete gastric juice containing pepsin and rennin for digestion of proteins, hydrochloric acid to provide acidic medium for action of enzymes and kill bacteria.
4. It has cardiac sphincter to allow food into the stomach.
5. It has pyloric sphincter to retain food in the stomach for digestion.

C. DIGESTION IN THE DUODENUM.

- ✓ Food/chyme enters the duodenum from the stomach through the **pyloric sphincter**.
- ✓ In the duodenum, food is mixed with bile and pancreatic juice.
- ✓ The presence of food in the duodenum stimulates the duodenal walls to secrete:
 - a) **Secretin** into blood stream which stimulates the liver to secrete bile which is stored in the gall bladder.

b) **Cholecystokinin** into blood which:

- i. Stimulates the gall bladder to release the bile and
- ii. Stimulates the pancreas to secrete pancreatic juice.

Note; The pancreas secrete hormones and secretes digestive enzymes (hence it has endocrine and digestive roles).

- 102 ✓ Bile contains salts (sodium hydrogen carbonate, sodium glycocholate and sodium taurocholate) which:
- i. Neutralize the acidic chyme from the stomach.
 - ii. Provide suitable alkaline medium for pancreatic enzymes.
 - iii. Emulsify fats (break down fats into tiny fat droplets) to increase the surface area for digestion by lipases. This is called **emulsification**.

Note: Emulsification is not chemical but physical/mechanical because fats are not broken down to fatty acids and glycerol.

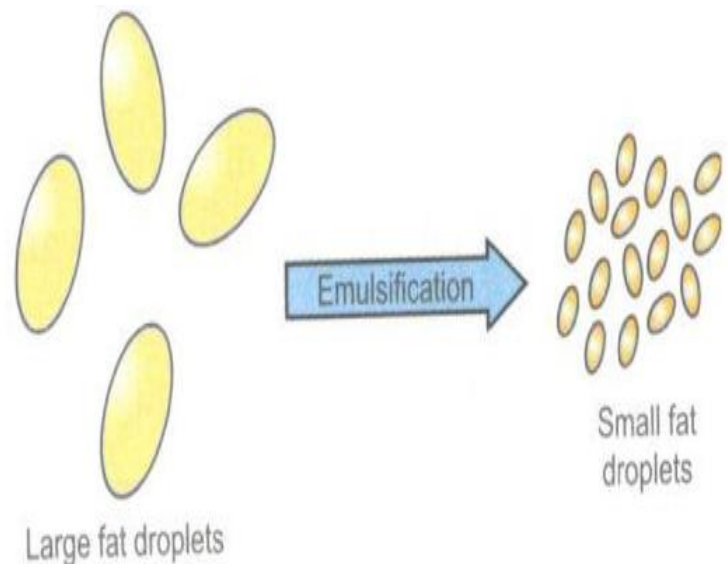


Fig 5.26: Illustration of emulsification.

✓ Pancreatic juice contains:

- ¹⁰³ 1. **Pancreatic lipase (enzyme)**-- which converts lipids to **fatty acids and glycerol**.
2. **Pancreatic amylase (enzyme)**-- which catalyzes the digestion/ conversion of remaining/ undigested **starch into maltose**.
3. **Sodium hydrogen carbonate (salt)** – which emulsifies fats, neutralizes the acidic chyme and provides alkaline medium suitable for action of pancreatic and intestinal enzymes.

4. **Trypsin (enzyme)**--which digests **proteins into peptides/ peptones**.
- ✓ **Trypsin** is secreted in inactive/ precursor form called **trypsinogen** to prevent the digestion of duodenal walls.
- ✓ It is converted into trypsin by an enzyme called **enterokinase**.

(Enterokinase enzyme)

Trypsinogen —————> **Trypsin**

Adaptation of duodenum.

1. It has Brunner's gland on its walls to secrete mucus for lubrication of food
2. It has crypts of Liberkuhn whose cells secrete digestive enzymes for digestion of food.
3. It is connected to the pancreas and the liver to supply pancreatic juice and bile respectively bile emulsifies fats/ lipids and neutralizes the acid from the stomach pancreatic juice contains enzymes for digestion of food.

PRACTICAL ACTIVITY.

Aim: To demonstrate emulsification of fats.

Requirements:

1. Sodium hydrogen carbonate solution.
2. Cooking oil.
3. Water.
4. Test tubes.
5. Ruler.
6. Measuring cylinder.

Procedure:

1. Pour 2 cm³ of cooking oil into the test tubes labelled A and B.
2. Add 2 cm³ of sodium hydrogen carbonate solution into test tube A. Rinse the measuring cylinder.
3. Add 2 cm³ of water into test tube B.
4. Shake the contents in both test tubes.
5. Write down your observations.

Observation.

- ✓ Formation of white emulsion.

¹⁰⁶ D) DIGESTION IN THE ILEUM.

- ✓ In the ileum, food is mixed with mucus and intestinal juice/ *succus entericus*.

Functions of the ileum

- i. To complete chemical breakdown of food.
 - ii. It provide a site for absorption of digested food into the blood.
- ✓ The inner walls of ileum has goblet cells which secrete mucus which;
 - i. Protects the wall of the intestines from being digested by protein digesting enzymes.
 - ii. Lubricates food/ allows smooth movement of food along the intestines.
 - ✓ The ileum walls also contain secretory/ epithelial cells (Cryts of liberkuhn) which secrete intestinal juice / *succus entericus*.

¹⁰⁷ ✓ Intestinal juice contains the following enzymes:

- a) **Maltase-** which catalyzes breakdown of maltose into glucose.
- b) **Sucrase/ invertase-** which catalyzes breakdown of sucrose into glucose and fructose.
- c) **Peptidase-** which catalyzes breakdown of peptides into amino acids.

- d) **Polypeptidase-** Which catalyzes the breakdown of polypeptides to amino acids.
- e) **Lipase-** which catalyzes breakdown of lipids (fats and oils) into fatty acids and glycerol
- f) **Lactase-** which catalyzes breakdown of lactase into glucose and galactose.

Note; Digestion of food is completed at ileum forming chyle. **Chyle** is food ready for absorption.

ADAPTATIONS OF SMALL **INTESTINES/ ILEUM.**

1. The small intestines are **long** to increase the surface area for absorption of food.
2. They are highly **coiled to slow down movement of food** to allow more time for digestion and absorption of digested food substances.
3. Inner lining has **numerous villi and micro-villi** to increase the surface area for absorption of food substances.
4. Their walls **have glands** (crypts of liberkurn) **that secrete intestinal juice which contains enzymes** for digestion of food.
5. Inner lining has **goblet cells** that secrete **mucus** to lubricate food and protect their walls against digestion by enzymes and lubrication of food.
6. It is **narrow/ has narrow lumen** to bring food into contact with intestinal walls/ blood vessels rapid/ faster absorption of food substances.

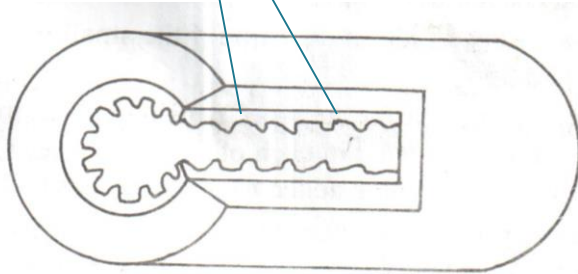
7. The walls have smooth muscles which contract and relax to facilitate peristalsis.
8. They are highly vascularized/well supplied with blood capillaries to create a steep/high concentration gradient hence faster diffusion of digested food/ for rapid and efficient removal of absorbed products.
9. They have lacteal vessels for absorption of fatty acids and glycerol.
10. They have thin epithelium to reduce the distance travelled by digested food hence faster absorption.
11. They have numerous mitochondria to provide energy for absorption of food.

ADAPTATION OF THE VILLUS.

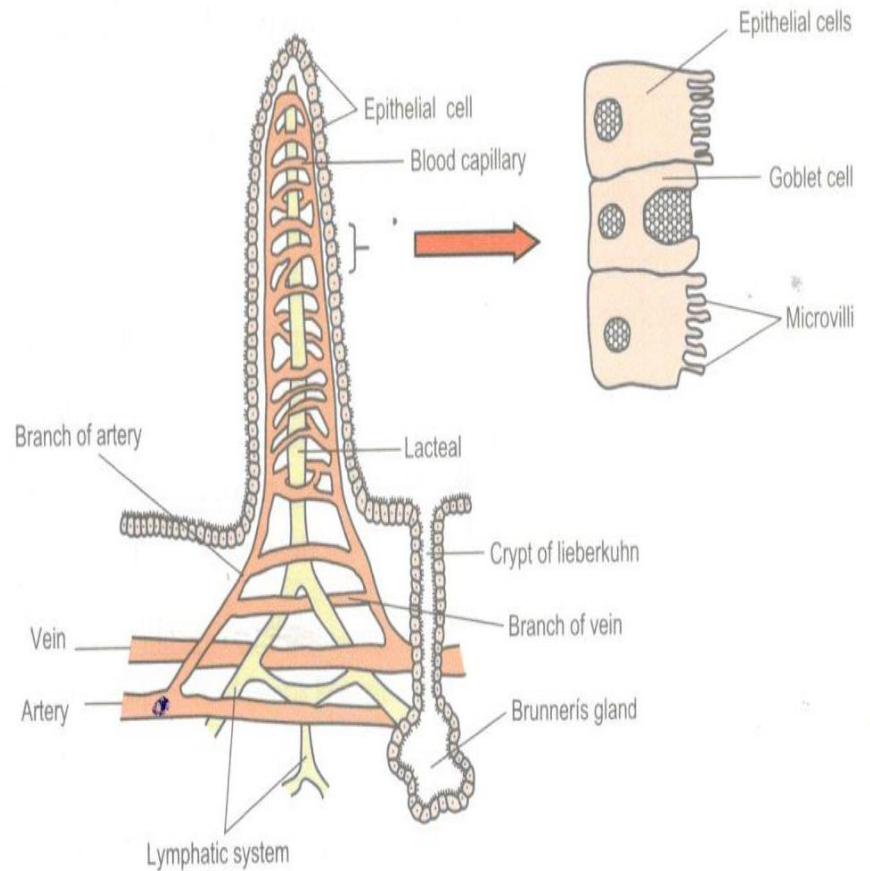
1. The villus has lacteal for absorption and transportation of fatty acids and glycerol.
2. It has a network of capillaries/highly vascularized to transport absorbed food substances.
3. It has thin epithelium to reduce the distance travelled by materials hence faster diffusion of digested food substances.
4. Has microvilli to increase the surface area for absorption of nutrients.
5. Has numerous mitochondria to provide energy for active transport of nutrients.

Structure of small intestine/ ileum

Villi



Structure of villus



ABSORPTION OF FOOD.

- ✓ **Absorption** is the process by which soluble products of digestion diffuse into the cellular lining of the villi into blood.
- ✓ **Glucose, fructose, galactose, Amino acids, mineral salts/ions, vitamins, Fatty acids and glycerol** are absorbed into the blood stream/capillaries through the epithelium of the villi by **diffusion and active transport**.
- The capillaries drain into the hepatic portal vein into the liver where amino acids and glucose are regulated before distributed to all body parts.
- ✓ Vitamins, mineral salts and water are absorbed directly without digestion.
- ✓ Water is absorbed through **osmosis**.

- ✓ Fatty acids and glycerol are absorbed into lacteals of the villi and combine to form **lipids** that pass into the lymphatic vessels and finally into the blood circulatory system.
- ✓ This distributes the fatty acids and glycerol to all parts of the body.
- ✓ The lacteals form the lymphatic system.

ASSIMILATION OF FOOD.

- ✓ **Assimilation** is the incorporation of absorbed food substances into cell metabolism.

- ✓ **Glucose** is oxidized to release energy during respiration and the excess is stored in the body as fats under the skin or in form of glycogen in the liver or muscles.
- ✓ **Fatty acids and glycerol** are oxidized to release energy, form new cell membranes and converted into fats and stored under the skin to insulate the body against heat loss and protect delicate internal organs.

- 114 ✓ **Amino acids-** are used to synthesize proteins for body growth, repair of worn out tissues and also oxidized to provide energy during starvation (in the absence of glucose and fats).
- ✓ Excess amino acids are excreted in the process called deamination.

FUNCTIONS OF THE COLON.

- i. Absorption of water and mineral salts.
- ii. Contain symbiotic bacteria which manufacture/ synthesize vitamin K, B₁, B₂, and B₁₂ and amino acids (which are absorbed into the blood stream).

Adaptation of colon.

1. Its walls consists of smooth muscles which contract and relax to facilitate peristalsis.
2. Inner lining has goblet cells which secrete mucus for lubrication.
3. Inner membrane is highly folded to increase surface area for absorption of water, mineral salts and attachment of micro-organisms.

ROLES/ FUNCTIONS OF CAECUM AND APPENDIX.

- ✓ The caecum and appendix in man have no functions but in herbivores they contain numerous symbiotic bacteria which secrete enzyme **cellulase** which digest cellulose into glucose.

ROLE OF ROUGHAGE IN DIGESTION.

- ✓ It is composed of cellulose and plant fibres.
- ✓ It adds bulk to food enabling it to have a grip/hold to the walls of the gut facilitating peristalsis allowing smooth movement of food in the alimentary canal.
- ✓ Lack of roughage in diet leads to constipation characterized by egestion of hard faeces.
- ✓ This is because food stays long in the gut and a lot of water is reabsorbed.

EGESTION

- ✓ This is the removal of undigested and indigestible food substances from the body (through the anus).

ROLE OF WATER IN DIET.

1. Acts as a solvent in which substances dissolve.
2. It acts as a medium for transportation of substances.
3. It acts as a medium in which metabolic reactions occur.
4. It facilitates hydrolysis of food substances.
5. It facilitates osmoregulation and bring about cooling effect in the body.

STUDY QUESTIONS

1. Describe digestion of fats and oils/lipids in humans

- ✓ In the duodenum, food is mixed with bile and pancreatic juice.
- ✓ Bile salts emulsify fats hence providing a large surface area for action of lipase enzymes and neutralizes the acidic chyme and provides alkaline medium for action of lipase enzyme.
- ✓ Pancreatic juice contains lipase which converts lipids into fatty acids and glycerol.
- ✓ In the ileum, food is mixed with intestinal juice secreted by intestinal wall. Intestinal juice contains lipase which converts/breaks down the remaining lipids into fatty acids and glycerol.

2. Describe digestion of proteins in humans.

- ✓ In the mouth, food is broken down mechanically by the teeth to increase the surface area for action of enzymes.
- ✓ The tongue rolls food into boluses and swallows into the stomach.
- ✓ In the stomach, food is mixed with gastric juice which contains Rennin and Pepsinogen.
- ✓ Pepsinogen is activated into Pepsin by Hydrochloric acid.
- ✓ Rennin (which is abundant in young children) converts the milk protein (Caseinogen) into Casein increasing its surface area for digestion by pepsin.
- ✓ Pepsin converts protein into peptides.
- ✓ In the duodenum, food is mixed with pancreatic juice which contains Trypsin which digests proteins into peptides.

- ✓ In the ileum food is mixed with intestinal juice (*Succus entericus*) secreted by intestinal walls.
- ✓ Intestinal juice contains peptidase enzyme which converts/ breaks down peptides into amino acids, polypeptidase enzyme which breaks down polypeptides into amino acids.

3. **Describe digestion of carbohydrates in humans.**

- ✓ In the mouth food is chewed/ mechanically broken down to increase the surface area for enzyme activity.

- ✓ Saliva contains salivary amylase/ ptyalin which acts on starch/ amylose and converts it into maltose. Saliva mixes with food and provides an alkaline medium for action of amylase.
- ✓ The tongue rolls food into boluses for swallowing into the stomach through peristalsis.
- ✓ In the duodenum, food is mixed with bile and pancreatic juice.

- ✓ Bile salt/sodium hydrogen carbonate provides alkaline medium for activity of duodenal enzymes and neutralizes acidic chyme/ food from the stomach.
- ✓ Pancreatic juice contains pancreatic amylase which converts starch to maltose.
- ✓ In the ileum epithelial cells in the ileum secrete succus entericus/intestinal juice which contains enzymes sucrase/invertase which converts sucrose to fructose and glucose, lactase which converts lactose to galactose and glucose and maltase which converts maltose to glucose.

FACTORS AFFECTING ENERGY REQUIREMENTS IN MAN

1. **Age**-Young children are actively growing/have many actively dividing cells/are physically more active hence require more energy than adults.
2. **Sex**- Males need more energy than females because they are more muscular hence have more cells that respire.

3. **Body size-** A small bodied person requires more energy because they have large surface area to volume ratio thus lose more energy than a person with big body. Some big bodied people may eat a lot of food to maintain energy demands of their large bodies.
4. **Activity/occupation-** A manual worker requires more energy than a sedentary worker.
5. **Lactation and pregnancy-** Pregnant mothers need more energy for foetal development while lactating mothers require more energy for milk production.
6. **Basal metabolic rate (BMR)-** this is the energy required by the body when it is at rest or the minimum amount of energy man requires to keep the body cells alive. The higher the BMR the higher the energy requirement.

ROLE OF VITAMINS IN THE BODY OF MAN/ HUMAN BEINGS.

VITAMIN	SOURCES	USES/ ROLE IN THE BODY	DEFICIENCY SYMPTOMS.
Vitamin A (retinal)	Liver, milk, eggs, carrots, fresh green vegetables,	✓ Night vision. ✓ Protects the skin and cornea from drying and becoming scaly.	✓ Poor night vision. ✓ Sore eyes. ✓ Dry and scaly skin and cornea. ✓ Poor resistance to diseases. ✓ Cold and bronchitis diseases.
Vitamin B1 (Thiamine)	Groundnuts, beans, whole cereals, egg yolk, milk, liver and kidney.	✓ Cell respiration. ✓ Proper growth in children.	✓ General weakness. ✓ Retarded growth in children. ✓ Beriberi disease. <u>Symptoms of beriberi disease</u> ✓ Limb paralysis. ✓ Heart failure. ✓ Swelling of legs (oedema). ✓ Loss of appetite/ diarrhoea/ vomiting. ✓ Weight loss/ muscle wasting. ✓ Pale skin.
Vitamin B2 (riboflavin)	Groundnuts, beans, whole cereals, egg yolk, milk, liver and kidney.	✓ Proper functioning of skin. ✓ Cell respiration.	✓ Pellagra disease. <u>Symptoms of pellagra disease</u> - Skin disorders. - Sores and bleeding in the mouth and gum.

VITAMIN	SOURCES	USES/ ROLE IN THE BODY	DEFICIENCY SYMPTOMS.
Vitamin B5 (Pantothenic acid)	Groundnuts, beans, whole cereals, egg yolk, milk, liver and kidney.	✓ Respiration. ✓ Proper functioning of the nervous system and alimentary canal.	✓ Malfunctioning of the nervous and alimentary canal.
Vitamin B12 (Cobalamine)	Liver, beef and kidney.	✓ Formation of blood cells	✓ Anaemia.
Vitamin C (Ascorbic acid)	Fresh fruits, green vegetables.	✓ Enhances absorption of iron. ✓ Promotes healing of wounds. ✓ Prevents infection/provides immunity against diseases. ✓ Helps in synthesis of connective tissues. ✓ Helps in development of healthy gums.	✓ Scurvy characterized by: a) Bleeding of gums. b) Anaemia. c) Swelling on the skin. d) Poor healing of wounds. e) Reduced/ poor resistance of diseases. f) Degeneration of muscles.
Vitamin D (Calciferol)	- It is a fat soluble vitamin manufactured in human body in the skin Sources are: Milk, fish, liver, egg yolk,	✓ Formation and hardening of bones. ✓ Strong teeth formation. ✓ Absorption of calcium and phosphorus.	✓ Rickets characterized by: a) Abnormal bone formation in children. b) Soft and brittle bones in adults.
Vitamin E	Milk, egg yolk, green	✓ Cell metabolism	✓ Infertility in some animals

PRACTICAL ACTIVITY.

Aim- Testing for vitamin C/
Ascorbic acid.

Requirements.

- a) Food substance in solution form.
- b) 0.1% Dichlorophenol indophenol (DCPIP)
- c) Test tube.
- d) Dropper.
- e) 10 ml measuring cylinder.

Procedure

1. Into a clean test tube put 1ml of DCPIP.
2. Add the food substance drop wise into the DCPIP in the test tube and shake well after each drop.

Observation

- ✓ DCPIP is decolorized i.e. it becomes colorless.

Conclusion.

- ✓ Vitamin C/ Ascorbic acid present.

Note- If the **blue** color of DCPIP is retained then vitamin C/ ascorbic acid is absent

ROLE OF MINERAL SALTS / ELEMENTS IN THE HUMAN BODY

ELEMENT	SOURCE	FUNCTIONS IN THE BODY	DISEASE CAUSED BY DEFICIENCY
Nitrogen	Meat, milk, eggs, fish	✓ Protein synthesis. ✓ Formation of cells, tissues, hair and nails.	✓ Kwashiorkor. ✓ Stunted growth.
Phosphorus	Protein foods	✓ Synthesis of proteins. ✓ Formation of bones and teeth. ✓ Formation of ATP.	✓ Rickets (characterized by poorly developed bones)
Calcium	Green vegetables, milk and cheese.	✓ Blood clotting. ✓ Muscle contraction. ✓ Formation of bones and teeth.	✓ Rickets ✓ Muscle cramps
Iodine	Iodinated salts, sea fish, cheese	✓ Formation of the hormone thyroxine	✓ Goitre (characterized by swelling of thyroid glands in the neck region.

ELEMENT	SOURCE	FUNCTIONS IN THE BODY	DISEASE CAUSED BY DEFICIENCY
Potassium	Liver, beef, vegetables, milk, eggs	✓ Transmission of nerve impulses. ✓ Contraction of muscles. ✓ Ionic/ osmotic balance.	✓ Muscular weaknesses ✓ Paralysis ✓ Nausea
Iron	Liver, eggs, green vegetables	✓ Formation of haemoglobin in red blood cells. ✓ Used in respiration.	✓ Anaemia. ✓ Low resistance to diseases/ low immunity.
Sodium	Table salt, fish, milk, green vegetables	✓ Maintains osmotic balance of body. ✓ Transmission of nerve impulses	✓ Reduced appetite ✓ Muscle cramps
Chlorine	Table salt	✓ Maintains osmotic balance of body. ✓ Transmission of nerve impulses	✓ Reduced appetite ✓ Muscle cramps

ELEMENT	SOURCE	FUNCTIONS IN THE BODY	DISEASE CAUSED BY DEFICIENCY
Sulphur	Protein foods	✓ Protein synthesis ✓ Formation of body tissues	✓ Kwashiorkor. ✓ Stunted growth.
Zinc	Liver, fish	✓ Activates enzymes	
Magesium and calcium	Meat, green vegetables	✓ Activates enzymes. ✓ Teeth and bone formation	✓ Nervous system disturbances
Manganese	Liver, kidney, tea, coffee, nut, vegetables, fruits	✓ Activates enzymes. ✓ Teeth and bone formation	✓ Abnormal bone and cartilage development

Practical activity 1.

Aim: To investigate the effect of temperature on enzyme activities.

Requirements:

- ✓ Six clean test-tubes.
- ✓ White tile.
- ✓ Test-tube holder.
- ✓ A dropper.
- ✓ Distilled water.
- ✓ Benedict's solution.
- ✓ Iodine solution.
- ✓ Water bath maintained at 37°C and 50°C.
- ✓ Source of heat.
- ✓ Thermometer.
- ✓ 10 ml measuring cylinder.
- ✓ 5 Labels.
- ✓ 6 cm³ of amylase/diastase solution.
- ✓ 10 cm³ soluble starch powder.

Procedure:

1. Label three test-tubes A, B and C.
2. Into test-tube A, B and C, add 3 cm³ of starch solution.
3. Into test-tube A, B and C add 2 cm³ of amylase/diastase solution.
4. Place test-tube C in the water bath maintained at 50°C for 30 minutes.
5. Place the remaining test tubes A and B in the water bath maintained at 38°C for 30 minutes.
6. After the 30 minutes, test for starch and reducing sugars on the contents of each test tube.
7. After the 30 minutes, test for starch and reducing sugars on the contents of each test-tube.

QUESTIONS

1. Which of the test tubes showed presence of starch?
 - *Test tube C.*
2. Which of the test tubes showed presence of reducing sugar?
 - *Test tube A and B.*
3. Explain the results.
 - *In test tube C, the enzyme was denatured by boiling hence starch was not hydrolyzed/broken down.*
 - *In test tube A and B, the enzyme amylase/diastase digested starch into maltose (a complex reducing sugar).*
4. What was the reason for maintaining the test tubes at 38°C?
 - *This is the optimum temperature for enzyme amylase.*
5. What was the reason for maintaining the test tubes at 50°C?
 - *At 50°C, the enzyme is denatured.*

PRACTICAL ACTIVITY 2.

Aim: To investigate the effect of pH on enzyme activity.

Requirements:

- ✓ Water bath kept at 37°C.
- ✓ Egg albumen suspension.
- ✓ 2m of hydrochloric acid.
- ✓ 2m sodium hydroxide,
- ✓ 3 test-tubes.
- ✓ 5 cm³ of 1% pepsin solution.
- ✓ Means of heating.
- ✓ Measuring cylinder.
- ✓ 5 Labels.

Procedure:

1. Label the test tubes A, B and C
2. Using a measuring cylinder, place 2 cm³ of the egg albumen suspension into each of the three test-tubes labeled A, B and C.
3. Add 1 cm³ of the 1 % pepsin solution to each of the three test tubes A, B, C.
4. To the contents of test-tube A, add three drops of the 2M hydrochloric acid, to B add three drops of distilled water and to C add three drops of the 2M sodium hydroxide solution.
5. Incubate the test tubes for 10 minutes in a water bath kept at 37°C.
6. Examine the test-tubes every two minutes, noting the cloudiness of the contents.
7. After the 10 minutes remove the test tubes and place them in a test-tube rack.

Test - tube	Observation of cloudiness					
	0Mi ns	2Mi ns	4Mi ns	6Mi ns	8Mi ns	10Mi ns
A						
B						
C						

Questions

1. Name the tube whose content has different appearance after ten minutes.
 - *Test tube A.*
2. Explain the change.
 - *The acid added provided the optimum pH for the enzyme hence the protein was digested.*
3. Which aspect of the enzyme properties does the experiment investigate?
 - *The optimum pH of the enzyme.*

PRACTICAL ACTIVITY 3.

Aim: To investigate the presence of catalase enzyme in living tissues.

Introduction.

- ✓ Catalase is an enzyme found in living tissues of plants and animals.
- ✓ It breaks down hydrogen peroxide (H_2O_2) produced during cellular metabolism into oxygen and water which are less toxic substances.
- ✓ Hydrogen peroxide is highly toxic hence it should not be allowed to accumulate in the tissues.
- ✓ If it were allowed to accumulate in the tissues, it would interfere with cellular metabolism.

Requirements:

1. Irish Potato.
2. Fresh piece of liver
3. Hydrogen peroxide.
4. 4 test-tubes.
5. Wooden splints.
6. Means of heating.
7. Scalpel blade.
8. Measuring cylinder (10 cm^3).
9. Labels.
10. Pair of forceps

Procedure:

1. Label four test-tubes A, B, C and D.
2. Measure 2 cm³ of hydrogen peroxide and put in the test-tubes A, B, C and D.
3. Cut a small piece of the Irish potato, place it in test-tube A and record your observation
4. Immediately, introduce a glowing splint into the mouth of the test tube and record your observation in the table below.
5. Cut a small piece of the fresh liver, place it in test-tube B and record your observation in the table below.
6. Immediately, introduce a glowing splint into the mouth of the test tube B and record your observation in the table below.
7. Cut a small piece of Irish potato and a small piece of liver and boil them for five minutes.
8. Remove a boiled Irish potato from hot water using a pair of forceps and place it in test tube C and record your observation in the table below.
9. Remove a boiled liver from hot water using a pair of forceps and place it in test tube D and record your observation in the table below.

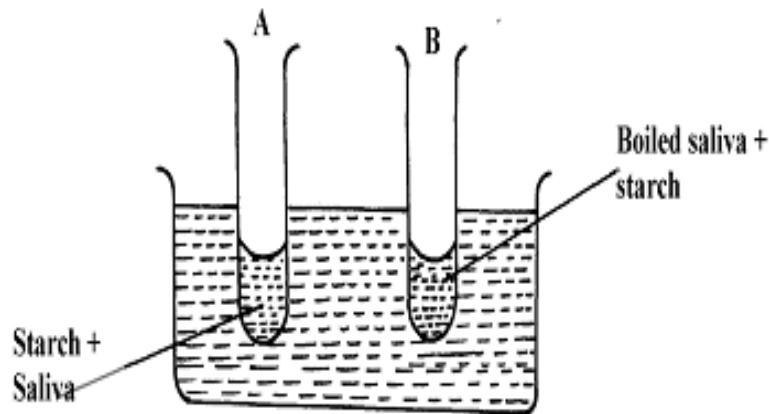
Questions

Test tube	Observation
A	- Bubbling/ effervescence. - Relighting of glowing splint
B	- Bubbling/ effervescence. - Relighting of glowing splint
C	- No bubbling/ effervescence
D	- No bubbling/ effervescence

- Account for the observation made in test tubes A and C.
 - ✓ *Test tube A- enzyme catalase present in tissue cells breaks down hydrogen peroxide into water and oxygen hence bubbling and the gas relights a glowing splint.*
 - ✓ *Test-tube B- boiling denatures catalase enzyme hence it did not break down hydrogen peroxide into water and oxygen.*
- Why were Irish potato and liver used?
 - ✓ *To show that catalase enzyme is found in both animal and plant tissues.*
- Which aspect of enzyme property does the experiment investigate?
 - ✓ *Sensitivity to changes in temperature.*

STUDY QUESTION.

- In an experiment to investigate on aspect of digestion, two test tubes A and B were set-up as shown in the diagram below. The test tubes were left in the bath for 30 minutes. The content of each test tube was then tested for starch using iodine solution.



- (a) What was the aim of the experiment?
 - ✓ *To investigate the effect of boiled saliva on starch OR to show the effect boiled/denatured enzyme amylase has on starch.*
- b) What results were expected in test-tube A and B.
 - ✓ *A- brown colour / colour of iodine persists / no change in colour.*
 - ✓ *B- blue black colour seen / colour changed to blue black.*
- c) Account of the results you have given in (b) above in test tube A and B.
 - ✓ *A- Starch has been digested / broken down / hydrolysed by salivary amylase hence no colour change.*
 - ✓ *B- Enzymes / Amylase denatured hence starch is not digested / broken down / hydrolysed.*